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Implementation of an Integrated Transformer for a Two-phase Interleaved Flyback PFC Rectifier

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Abstract

Universal Serial Bus Power Delivery (USB PD 3.1) is a widely-applied standard for a lot of electronic device charging. In 2021, USB PD 3.1 was announced with a variable output of 5 V to 48 V and a maximum output power of 240 W. A multi-phase flyback rectifier is a simple circuit capable of power factor correction, galvanic isolation, and wide output voltage range, and is thus preferred among other topologies for USB PD. However, the number of magnetic components increases with the number of phases and the power density decreases. This paper proposes the design of an integrated transformer for a two-phase interleaved flyback rectifier to reduce the size and number of magnetic components. The independent operation of the integrated transformer in two phases was verified using a 240-W two-phase interleaved flyback rectifier at 100 kHz switching frequency operating at discontinuous conduction mode.

Contents

List of Figures

List of Tables

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I also dedicate this work to my family - Mama, Papa, Kuya, Boy, Ate Ruby and Achie - who have always been there to listen to all my struggles even if they do not understand a thing at all in my work. They have helped me to keep going until today.

This is also for the Filipino People, who have motivated me to pursue higher studies for their betterment in the future.

And of course, this is also for God, for helping me and guiding me all throughout my study and helping me reach where I am today.

Chapter 1

Introduction

1.1 Power Electronics

Power Electronics is a discipline of Electrical Engineering that deals with conversion of power from one form to another. The main goal of

A converter is an electrical circuit designed to change an input voltage into a different output voltage. Depending on the application, both the voltage level and the kind of voltage being outputted (i.e. DC or AC) may be manipulated.

There are four common kinds of converters commonly implemented today. These include: DC-DC Converters, AC-DC Converters (Rectifiers), DC-AC Converters (Inverters), and AC-AC Converters.

DC-DC Converters convert a DC input to a DC output, and they can either raise up the voltage level (boost type or step-up DC-DC Converter), or bring down a voltage level (buck type or step-down DC-DC Converter). A buck converter is commonly used in various appliances that we own today. For instance, for charging, once AC is converted to DC for the battery, some components for our appliances need lower level DC to operate safely. Buck converters can be used here. LED Lighting systems commonly use Buck Converters to run efficiently. It is also used a lot in automotive applications, especially for powering small electrical components that require only a small amount of voltage to operate. A boost converter is also

used commonly in battery-powered devices. For instance, some components in our automobiles, or laptops, or smartphones require higher voltage to operate.

DC-AC Converters convert a DC input to an AC output of desired voltage and frequency level and are commonly known as inverters. They commonly utilize PWM Control to be able to convert voltage. Their most common applications include the automotive industry, especially for powering air conditioners, and electric motors for electric vehicles. During power outages or camping excursion for example, portable generators are often used to power various appliances, and in order to do so, they convert DC to AC first, which then go into the appliances for their own conversion to DC and finally to be used. Lastly, they are also used a lot of times in industrial application, especially for controlling motors, and in electro-welding and electro-forming processes.

AC-DC Converters convert an AC input to a DC output, and are also commonly known as rectifiers. These are commonly used in charging systems and power supplies too, as had been explained, and they take care of the part of converting AC into usable DC by our devices. It is also commonly used in some renewable power systems, such as solar power systems, to convert AC from the solar panel to DC power stored in batteries, which can then be used to power various devices.

AC-AC Converters convert an AC input to an AC input (to a different level) also, and are also commonly known as transformers. One of the most common applications is power distribution from an electrical grid. Power lines from power plants carry very high level AC, and for it to be usable in a home setting, this AC must be converted to a different level and frequency for it to be usable in our appliances.

This study focuses on the circuit for USB Power Delivery, which is an example of such application is electric charging, which makes use of power conversion from power supply to a load. This is

Table 1.1: Caption

1.2 USB Power Delivery

USB Power Delivery (PD) is a widely-applied charging technique today. It is widely used to promote fast charging for instance for laptops, devices, and other everyday household appliances. A summary of the specifications of recent PD Standards are summarized in Table 1.1.

The latest standard is called USB PD 3.1, which allows for up to 240 W of output power across a wide output voltage range. To implement USB PD 3.1, the following standards are required: (1) to allow for galvanic isolation as a measure to protect the user of the device from circuit failure, (2) to have

1.3 Topologies for implementing USB Power Delivery 3.1

There are a lot of different topologies available today to implement USB Power Delivery. This section will introduce some of their advantages and disadvantages.

1.3.1 PFC Boost, LLC, and Buck Converter Combination Circuit

1.3.2 PFC Boost with AHB Flyback

1.3.3 Conventional PFC Flyback

1.3.4 Two-phase Interleaved Flyback PFC Rectifier

To solve the problem of design complexity, component count, and load restriction, interleaving is proposed. A two-phase interleaved flyback PFC rectifier is examined

in this study, as shown in Figure 1.1, with the circuit operating in DCM, and two-phase drive with 180° phase difference. Its operating principle will be discussed in detail in Chapter 3. An additional benefit of interleaving is to reduce current stresses on the switches, since half of the total power delivered to the load is transferred from the primary side to the secondary side in each phase. This then allows the circuit to accommodate higher load conversion.

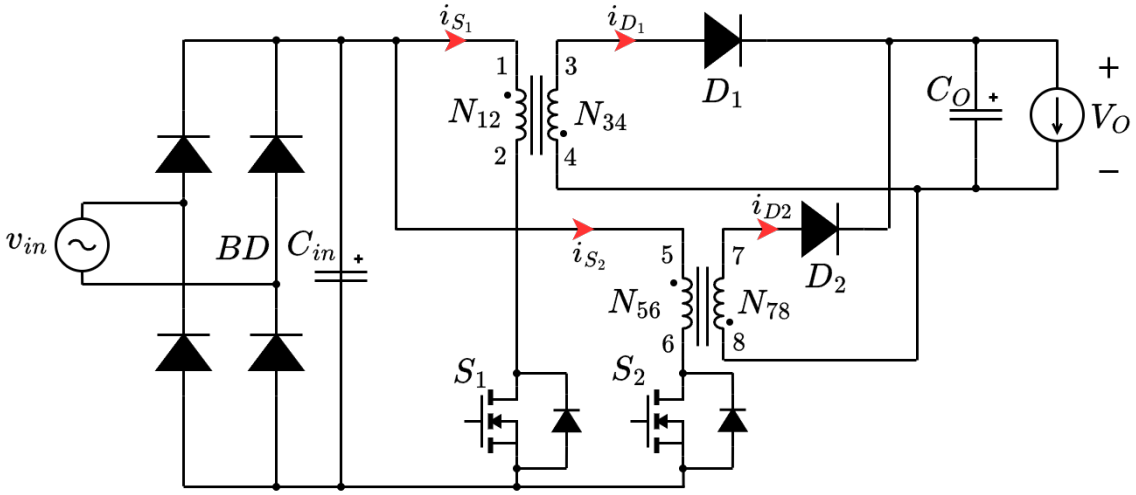


Figure 1.1: A two-phase interleaved flyback PFC rectifier.

However, there is still the problem of the magnetic component of the circuit being too large, leading to low power density. To solve this problem, an integrated transformer is proposed that can operate separately in two phases due to elimination of coupling between the two transformers completely.

1.4 Concept of an Integrated Transformer for the Two-phase Interleaved Flyback PFC Rectifier

Miniaturization of the circuit is accomplished by a new integrated transformer that can operate separately in the two phases of the circuit operation. This is accomplished by the winding structure for the transformer as shown in Figure [12]-[13]. The number of turns between terminals i and j corresponds to N_{ij} in Figure. The integrated transformer has zero coupling coefficient coefficients between the two

transformers N_{12}, N_{34} and N_{56}, N_{78}. More details about the operational principle of the transformer will be shown in Chapter xx.

1.5 Objectives of the Study

In this study, a design method of such a transformer is presented. Moreover, a simulation is performed to confirm the design method. The miniaturization effect was examined through a volume comparison. An experimental verification for the transformer operation was performed using a prototype.

1.6 Significance of the Study

Chapter 2

An Overview of Integrated Magnetics

2.1 Integrated Magnetics

Integrated magnetics are a long existing technique to minaturize the magnetic components in a circuit. This section will detail some existing techniques that are available for integrated magnetics. As will be explained in this section, the goal is to reduce the number of magnetic cores for instance, as they are the component that can take up the most volume in terms of passive components. To be able to integrate multiple components into one, the winding structure should be carefully thought of.

To design an integrated magnetic component, one must take into account the following:

- **Circuit operational principle.** Circuits operate differently from each other. Some circuits may operate in for example, DCM or CCM. Some circuits may operate in step-up or step-down operation. Some circuits operate in different number of phases. Depending on these conditions, the transformer will conduct consecutively in the primary and secondary side. The flux interaction caused by the winding interaction must then be carefully taken into account

during the design process.

- **Target feature or improvement.** Some magnetic components are designed to improve certain features in the circuit. For instance, some magnetic components are designed to improve current ripple in the input and output side. Some magnetic components are designed to improve efficiency by controlling the current ripple in various parts of the circuit.

- **Flux operation.** The flux interaction in the transformer can affect the circuit behavior and the size requirement of the core used for magnetic integration. This should be carefully considered in design to optimize the core used for the transformer.

2.2 Common Integrated Magnetics Techniques

There is no integrated magnetic technique made exactly for the circuit being examined in this study. However, in this section, five studies are examined which employ an integrated transformer in two-phase driving. The focus of each study is different, and only some of the most significant contributions or changes are detailed in this review.

2.2.1 Integrated Transformer by Ishihara et. al. (2015)

The transformer was implemented in a two-phase interleaved forward converter, operating in continuous current mode (CCM), at a step-up and step-down operation. The proposed transformer by the authors is shown in Figure 2.1, which is the transformer implemented in parallel-input mode. The step-down operation will be reviewed in this section.

The authors examine the transformer in both series and parallel input connection. The transformer was connected in parallel at the output side. If one wants to

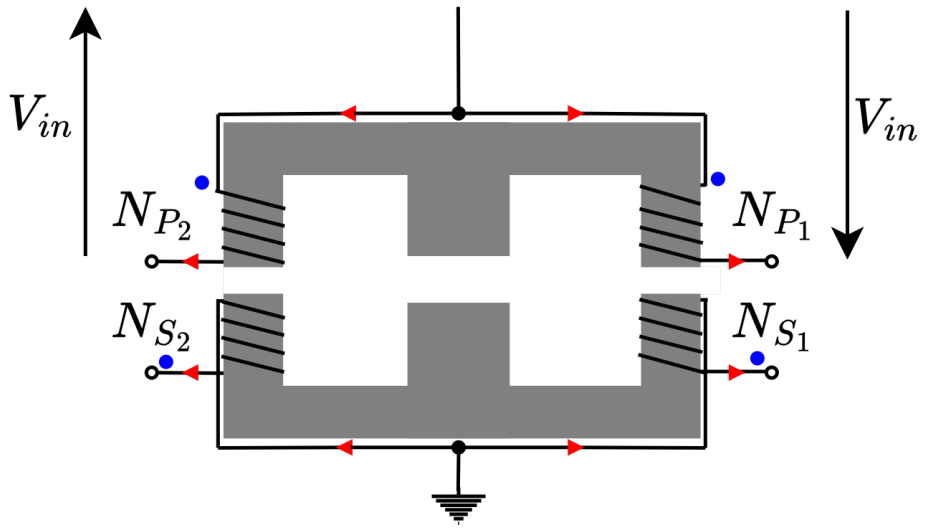


Figure 2.1: Transformer by Ishihara et. al. (2015).

reduce the voltage stress on the switching elements, the inputs must be connected in series, whereas to reduce the current stress, the inputs must be connected in parallel.

The primary and secondary side windings are placed on the outer legs only. The winding is designed such that no flux cancellation occurs in any of the legs. An air gap is placed in the center leg. The transformer was designed for 400 V − 24 V conversion at 200 W power output. Magnetic saturation does not occur in all the legs of the transformer. The study identified the input capacitor as the component with the largest loss and the transformer in this study was focused to reduce the size of the input capacitor. A thorough loss analysis was provided in this study. From this study, it was found that the series input configuration was more preferred more than the parallel configuration for the step-down condition, as it proved to have less losses and higher power density compared to the parallel configuration at the testing specifications. However, for this circuit and its transformer, the input voltage is very high compared to the input voltage used in this study. The loss reduction is not the focus of this study, and this study has a different focus compared to the study described here.

The design method uses the primary side current ripple and the coupling coefficient to decide the saturation magnetic flux density, to be able to decide the proper

core. This makes the design method more complicated than usual.

2.2.2 Integrated Transformer by Taheri et. al. (2011)

The transformer was implemented in a two-phase interleaved forward converter, operating in continuous current mode (CCM), at a step-down operation. The proposed transformer by the authors is shown in Figure 2.2. The proposed transformer and converter was not verified experimentally in this study. The transformer was connected in series in the input side and in parallel at the output side. This was done to reduce the voltage stress in the switching devices in the primary components. Moroever, because of the parallel connection in the secondary side, ripple current will be reduced in the output side. There is simultaneous conduction in the secondary side.

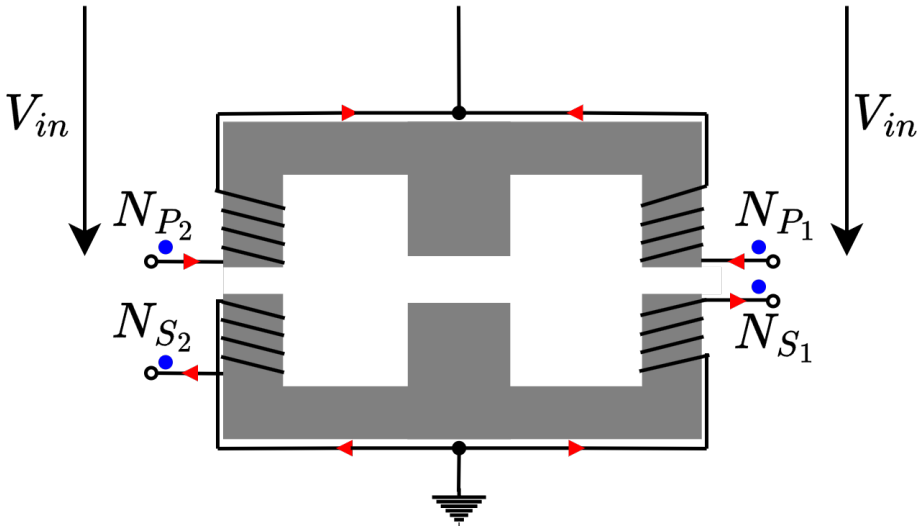


Figure 2.2: Transformer by Taheri et. al. (2011).

The primary and secondary side windings are wound on the outer legs only, and the winding makes it possible for flux in the center leg to be cancelled in the center leg, which can therefore reduce the amount of flux in the core over all, and can therefore contribute to less core loss. The windings are also implemented such that there is coupling between the primary and secondary side windings, which can somehow reduce the input current ripple, although the amount of coupling was not specified in the study, and moreover, the topology this transformer is being

implemented into is different from the one focused in this study.

The circuit used in this study enables ZVS, which is a well-known technique to improve efficiency and various circuit characteristics, and this then makes the design of the transformer more complicated, as the values of leakage inductance should be chosen and set accordingly with the switch capacitance. Moreover, the design method uses the primary side current ripple and the coupling coefficient to decide the saturation magnetic flux density, to be able to decide the proper core. This makes the design method more complicated than usual.

2.2.3 Integrated Transformer by Qian and Lehman (2007)

The transformer was implemented in a two-phase interleaved flyback converter, operating in continuous current mode (CCM), at a step-down operation, for high input voltages. The transformer by the authors is shown in Figure 2.3. The transformer was connected in series in the input side and in parallel at the output side. This was done to reduce the voltage stress in the switching devices in the primary components. There is simultaneous conduction in the secondary side. A lot of switching devices already have a high voltage rating. However, the current stresses will not be reduced in the switching devices, which may necessitate switching devices with higher current ratings and are more expensive. The parallel connection in the secondary side can still keep the benefit of ripple current in the output side.

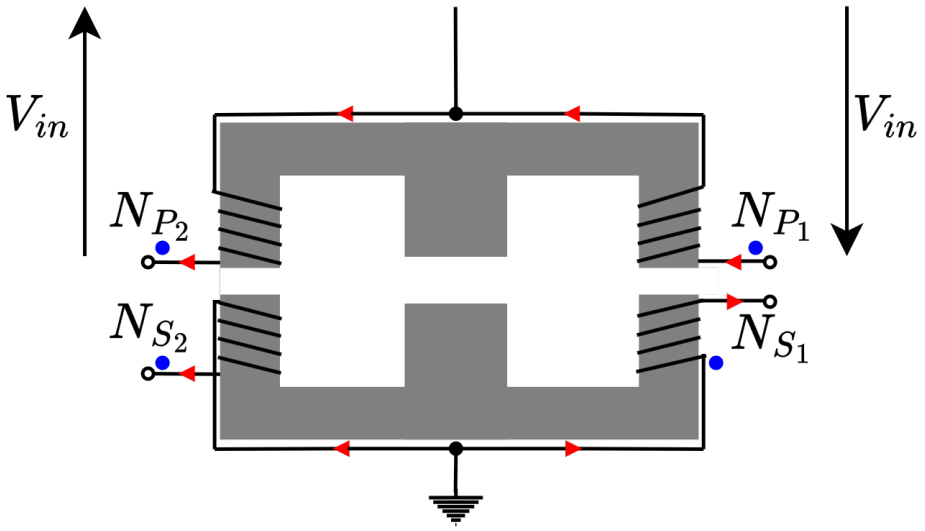


Figure 2.3: Transformer by Qian and Lehman (2007).

The primary and secondary side of the transformers were inversely coupled to each other. Moreover, all the legs were gapped, causing a weak coupling between the windings, and therefore contributing to the reduction of current spike caused by voltage mismatch. However, this coupling causes a large leakage inductance, and can cause problems in distortion and noise in the waveforms, as can be seen in the results of this study. This is mainly due to the fact that flux can go through all the

legs at all times. These points can be improved, although is not the main focus of this study.

The transformers are wound on the outer legs only, and the winding structure makes magnetic flux non-cancellable in the transformer at any time. The design method uses the primary side current ripple and the coupling coefficient to decide the saturation magnetic flux density, to be able to decide the proper core. This makes the design method more complicated than usual.

2.2.4 Integrated Magnetic Component by Yan et. al. (2003)

The transformer was implemented in a full-wave converter, operating in continuous current mode (CCM), at a step-down operation. There is simulatneous conduction in the primary and secondary side. The proposed transformer by the authors is shown in Figure 2.4. The transformer was connected in parallel at the input side and in parallel at the output side.

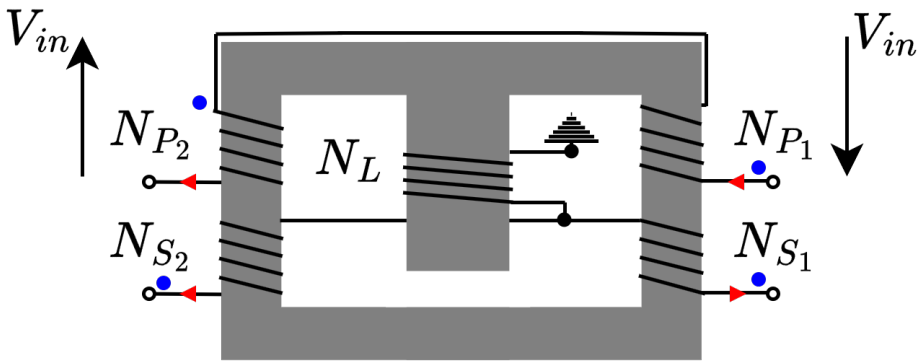


Figure 2.4: Transformer by Yan et. al. (2003).

Compared to the other integrated transformers introduced in this study, this transformer features an additional inductor winding at the center leg. The inductor winding was added to control the output ripple. Flux cancellation does not occur in the core also. Moreover, this transformer was implemented using a planar core, which can then improve flux distribution and help in core volume miniaturization. However, it is important to note that this integrated magnetic component was designed in low voltage specifications (75 V $-$ 3.3 V conversion), and can only support up to 100 W. The efficiency of the circuit is considerable at 92.9%.

The design method uses the primary side current ripple and saturation magnetic flux density, to be able to decide the proper core. No information about the coupling coefficient is given and seems to be not a large consideration in the study, although there seems to be coupling according to the results.

Chapter 3

Operational Analysis and Core Volume Analysis

This chapter will describe the operational principle of the two-phase interleaved flyback rectifier analyzed in this study with an integrated transformer.

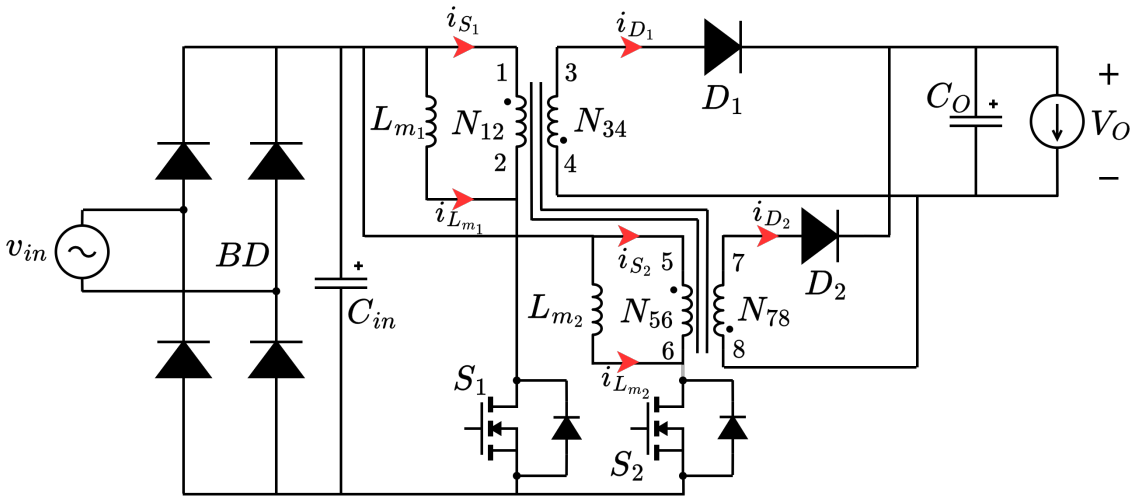


Figure 3.1: Proposed Two-phase Interleaved Flyback PFC Rectifier.

The proposed integrated transformer was designed for a two-phase interleaved flyback PFC rectifier operating in discontinuous conduction mode (DCM). Each switch S_1, S_2 in Figure 3.1 operates uniformly at a constant duty cycle with 180-degree phase shift.

To enable independent operation per phase without magnetic interference, any coupling between the two transformers is eliminated. The following winding structure makes it possible for the conventional, two-separate core structure to be integrated into one, single core, allowing for downsizing of magnetic components. Details of the influence of the winding structure on the operating principle will be explained further.

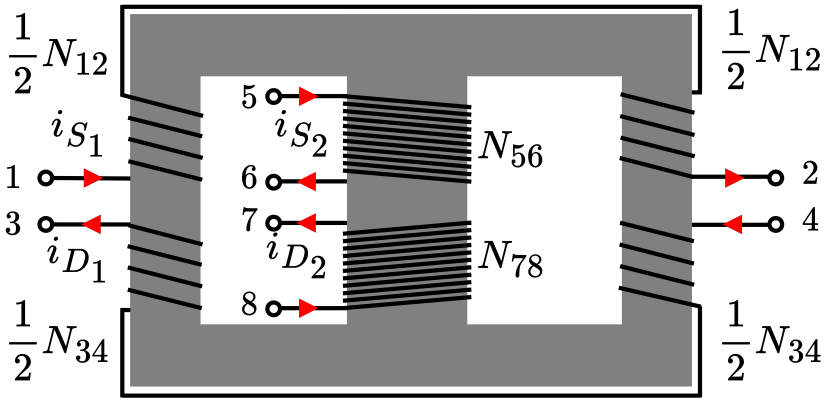


Figure 3.2: Proposed Integrated Transformer.

The winding structure is shown in 3.2. Half of the total number of turns for transformer N_{12} and N_{34} are wound at each outer leg with opposite polarity. Moreover, N_{56} and N_{78} are also wounded with opposite polarity at the center leg. As an additional note, the magnetic resistance of each outer leg is also made equal.

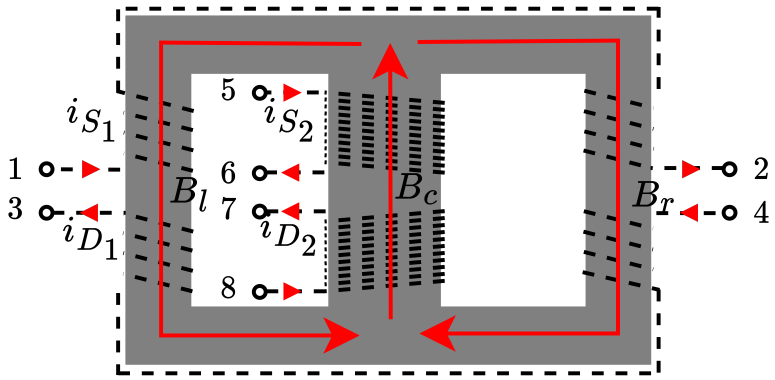
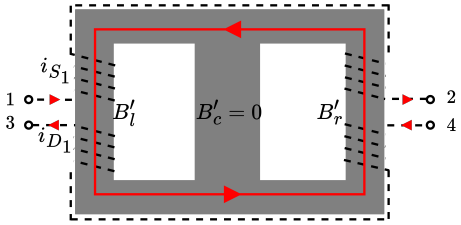
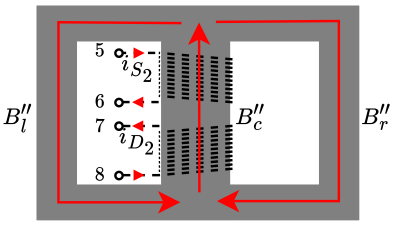


Figure 3.3: Flux densities inside the proposed integrated transformer. B_l represents the flux density in the left leg, B_r represents the flux density in the right leg, and B_c represents the flux density in the right leg.



(a) Outer transformer $N_{12}N_{34}$ with flux densities.



(b) Center leg transformer $N_{56}N_{78}$ with flux densities .

Figure 3.4: Integrated transformer separated according to AC operation cycle.

The behavior of various electrical quantities and also the flux density will be examined from now on. For the flux density in the core, especially that the transformer can operate separately in two phases, the principle of superposition can be used to examine the flux behavior in each leg of the transformer for each phase. This is important in deciding the saturation magnetic flux density, which is important for the design of the integrated transformer. From the transformer in Figure **??** B_l, B_r, and B_c are the magnetic flux densities in the left, right, and center legs respectively. The transformers can then be separated into each phase as per the principle of superposition. As shown in Figure **??** and **??** B'_l, B'_r, and B'_c represent the flux densities in the positive AC Cycle, while B''_l, B''_r, and B''_c are the flux densities in the negative AC Cycle. By the principle of superposition, shown in Equation **??** are the corresponding expressions for the superimposed flux densities in each of the legs.

$$\begin{cases} B_l &= B'_l + B''_l \\ B_r &= B'_r - B''_r \\ B_c &= B''_c \end{cases} \tag{3.1}$$

The following were considered in this analysis:

1. The cross-sectional area of the central leg in the EE core is twice that of an outer leg, as per commercial standard, causing N_{12} to have twice the number of turns of N_{56} and N_{34} to have twice the number of turns of N_{78}.

2. Secondly, the magnetizing current and magnetic flux density are proportional, with B_max' as the maximum flux density in the transformer at the maximum magnetizing current.

3. The saturation magnetic flux in each separate transformer during each phase is expected to be consistent. This is because the magnetizing inductance is set the same for each phase as well, and each phase is driven at the same conditions.

4. Lastly, to simplify the analysis, the resonant operation between the magnetizing inductance and the parasitic capacitance of the switch is omitted.

3.1 Circuit and Transformer Operational Analysis

Crucial in the operational analysis of the circuit is the discharge time of the transformer after the switches are turned off in the primary side. The duty cycle D influences how fast the transformer discharges, and will be explored in this section.

The circuit and transformer is first analyzed according to the condition where $D \leq 0.5$ and $D \leq 0.5\frac{nV_O}{\sqrt{2}V_{\text{in, max}}}$. In other words, one transformer discharges before the other fully charges, or its magnetizing reaches the peak.

A compilation of the essential waveforms involved in the circuit and transformer operation is shown in Figure **??**. v_{gs_1} and v_{gs_2} represent the corresponding gate-source voltages, i_{S_1} and i_{S_2} represent the corresponding switch currents, i_{D_1} and i_{D_2} represent the corresponding secondary side diode currents, $i_{L_{m_1}}$ and $i_{L_{m_2}}$ represent the corresponding magnetizing inductance currents. B_l, B_r, and B_c are the magnetic flux densities in the left, right, and center legs respectively. B_l', B_r', and B_c' represent the flux densities in the positive AC Cycle, while B_l'', B_r'', and B_c'' are the flux densities in the negative AC Cycle.

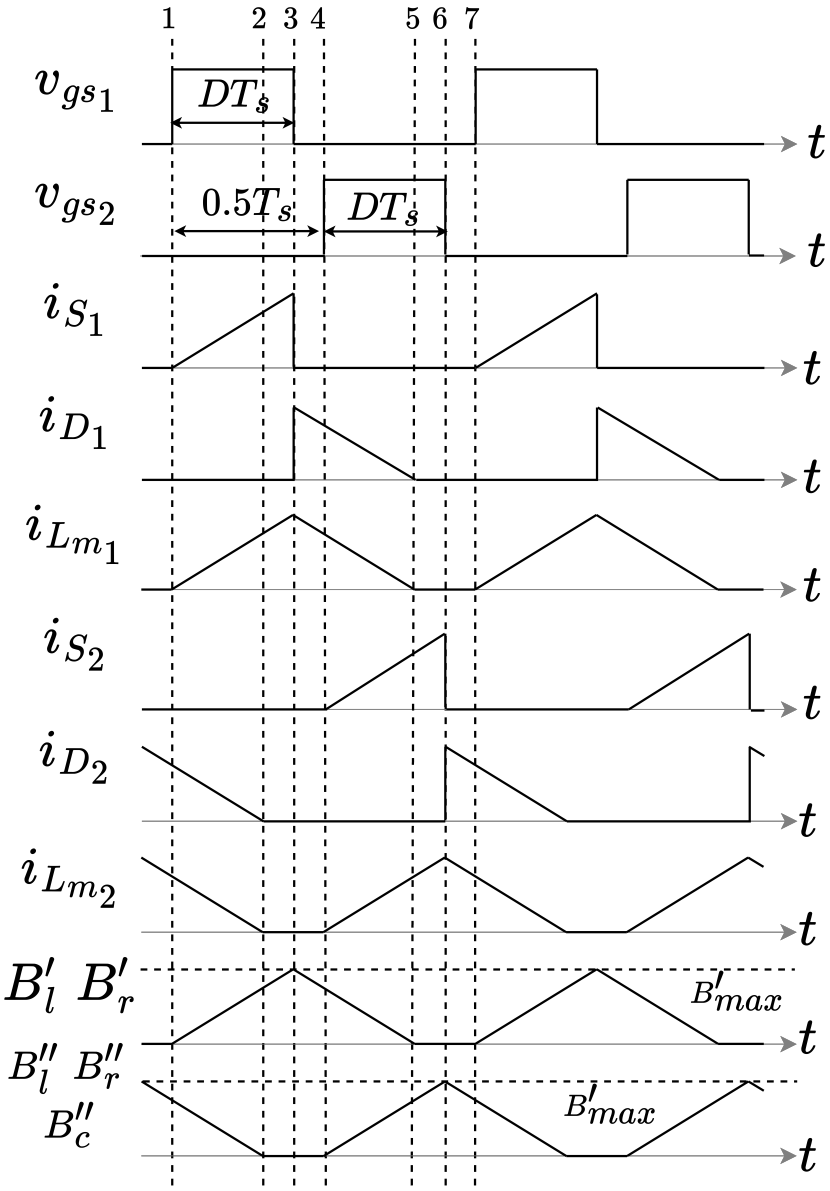


Figure 3.5: Key electrical and flux waveforms of the circuit when $D \leq 0.5$ and $D \leq 0.5\frac{nV_O}{\sqrt{2}V_{\text{in, max}}}$.

The circuit operates in the following manner in the positive AC cycle.

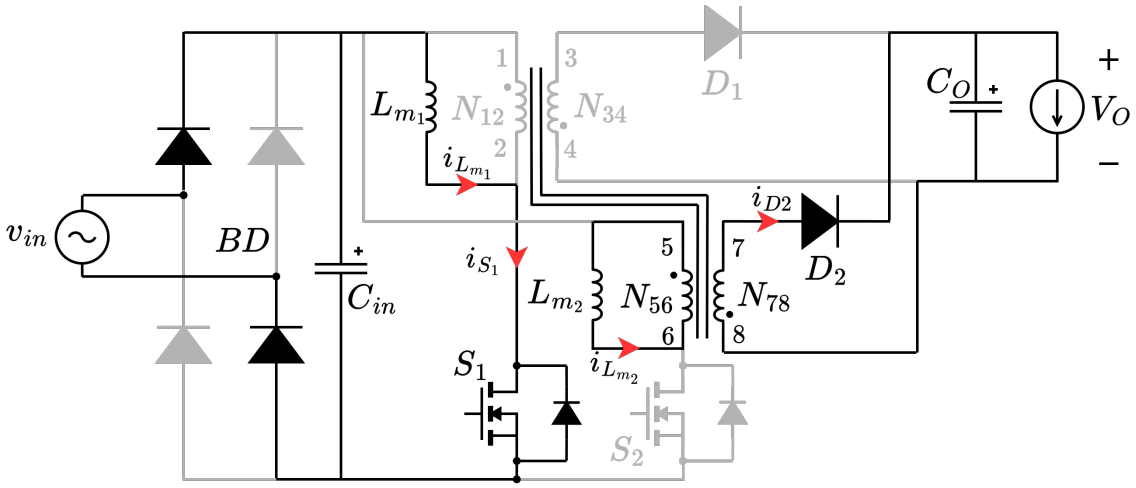


Figure 3.6: Operation of the proposed circuit with the transformer at Mode 1, $t_1 < t < t_2$

Mode 1, $t_1 < t < t_2$: With S_1 on and S_2 off, the following happens. Here, the transformer $N_{56}N_{78}$ continues to discharge until $i_{L_{m_2}}$ and i_{D_2} reaches 0. $i_{S_2} = 0$ since the corresponding switch S_2 is off. The fluxes B_l'', B_r'', and B_c'' decrease linearly as well until they reach 0. On the other hand, since transformer $N_{12}N_{34}$ is storing energy in its magnetizing inductance, then $i_{L_{m_1}}$ should be increasing linearly, and also B_l' and B_r'. i_{S_1} must be increasing linearly as well. Moreover, in this phase, since $N_{12}N_{34}$ is storing energy, then i_{D_1} must be 0. A representation of the active components in this mode are shown in Figure **??**

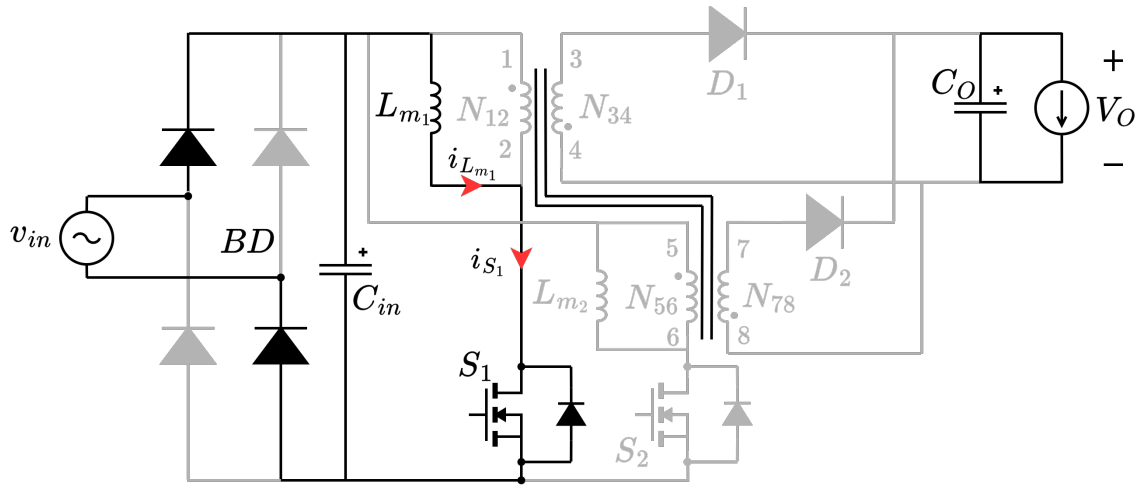


Figure 3.7: Operation of the proposed circuit with the transformer at Mode 2, $t_2 < t < t_3$

Mode 2, $t_2 < t < t_3$: B_l' and B_r', i_{S_1}, and $i_{L_{m_1}}$ reach their peak, as the magnetizing inductance L_{m_1} becomes fully charged, ready for energy transfer. Here, the quantities related to $N_{56}N_{78}$ stay at 0, since it is inactive during this time period. A representation of the active components in this mode are shown in Figure **??**

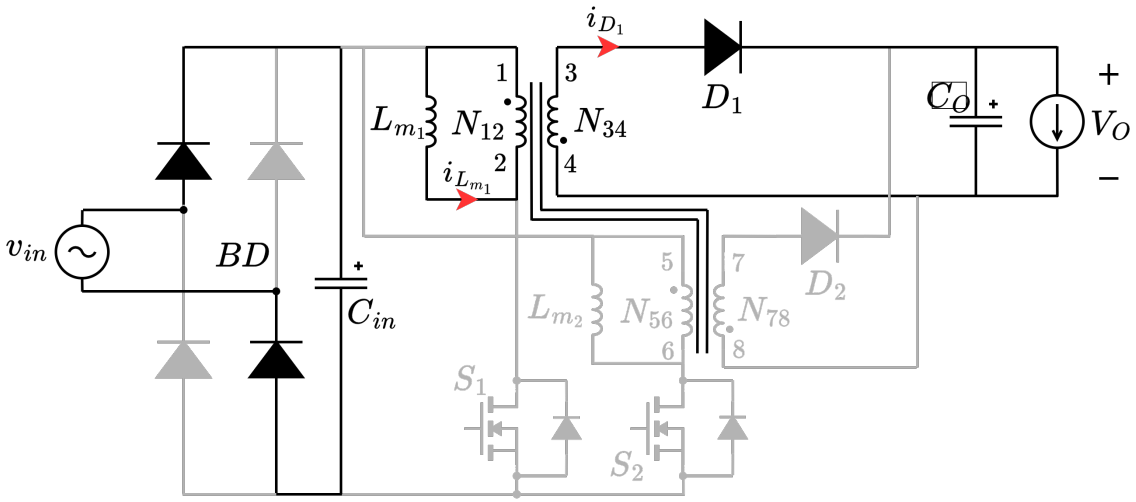


Figure 3.8: Operation of the proposed circuit with the transformer at Mode 3, $t_3 < t < t_4$

Mode 3, $t_3 < t < t_4$: S_1 turns off, signifying that $i_{L_{m_1}}$ has reached its peak value. The transformer $N_{12}N_{34}$ has finished charging and has started to discharge. This explains why i_{D_1}, $i_{L_{m_1}}$, and the flux densities B_l', and B_r' start decreasing linearly. The corresponding currents and flux densities to $N_{56}N_{78}$ stay at 0, since it is inactive during this time period and S_2 is still off. A representation of the active components in this mode are shown in Figure **??**

The circuit operates in the following manner in the negative AC cycle.

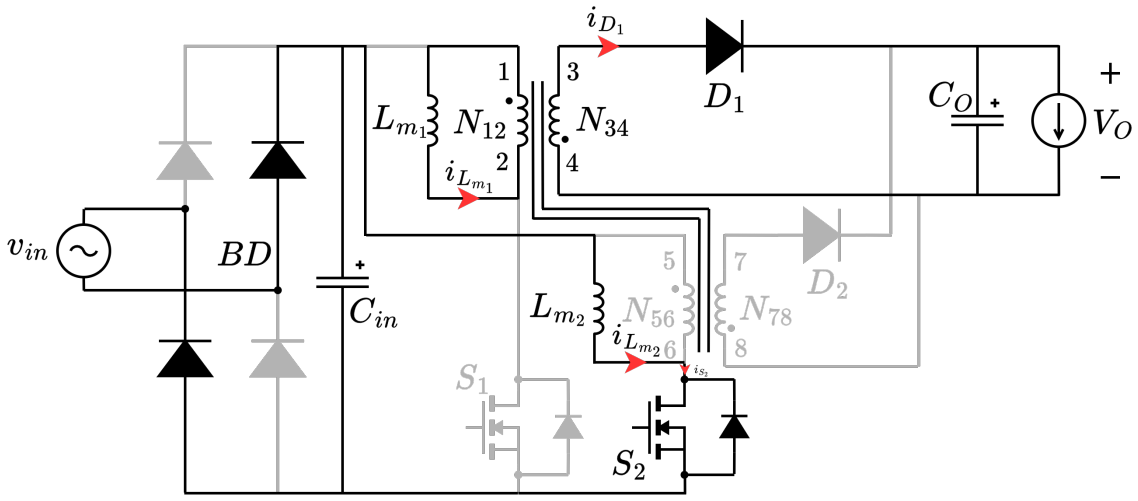


Figure 3.9: Operation of the proposed circuit with the transformer at Mode 4, $t_4 < t < t_5$

Mode 4, $t_4 < t < t_5$: Contrary to Mode 1, with S_2 on and S_1 off. The transformer $N_{12}N_{34}$ continues to discharge until its corresponding magnetizing inductance current $i_{L_{m_1}}$ and secondary diode current i_{D_1} reaches 0. $i_{S_1} = 0$ since S_1 is off. B'_l, and B'_r decrease linearly as well until they reach 0. On the other hand, since transformer $N_{56}N_{78}$ is storing energy in its magnetizing inductance, then $i_{L_{m_2}}$ should be linearly increasing, and also B''_c, B''_l and B''_r. i_{S_2} must be linearly increasing as well. Moreover, in this phase, since $N_{56}N_{78}$ is storing energy, then i_{D_2} must be 0. A representation of the active components in this mode are shown in Figure **??**.

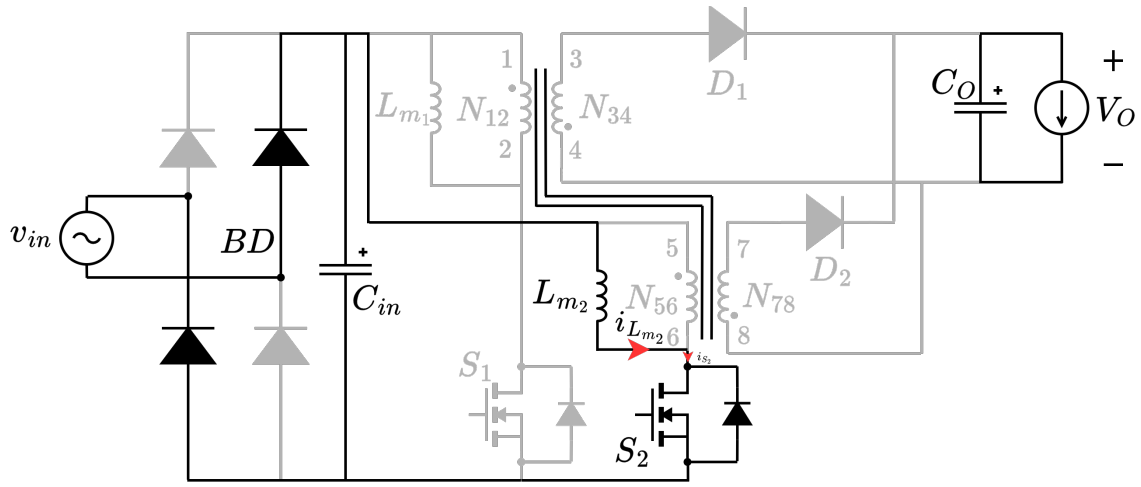


Figure 3.10: Operation of the proposed circuit with the transformer at Mode 5, $t_5 < t < t_6$

Mode 5, $t_5 < t < t_6$: B_l'', B_c'' and B_r'', i_{S_2}, and $i_{L_{m_2}}$ reach their peak, as the magnetizing inductance L_{m_2} becomes fully charged, ready for energy transfer. Here, the quantities related to $N_{12}N_{34}$ stay at 0, since it is inactive during this time period. A representation of the active components in this mode are shown in Figure **??**.

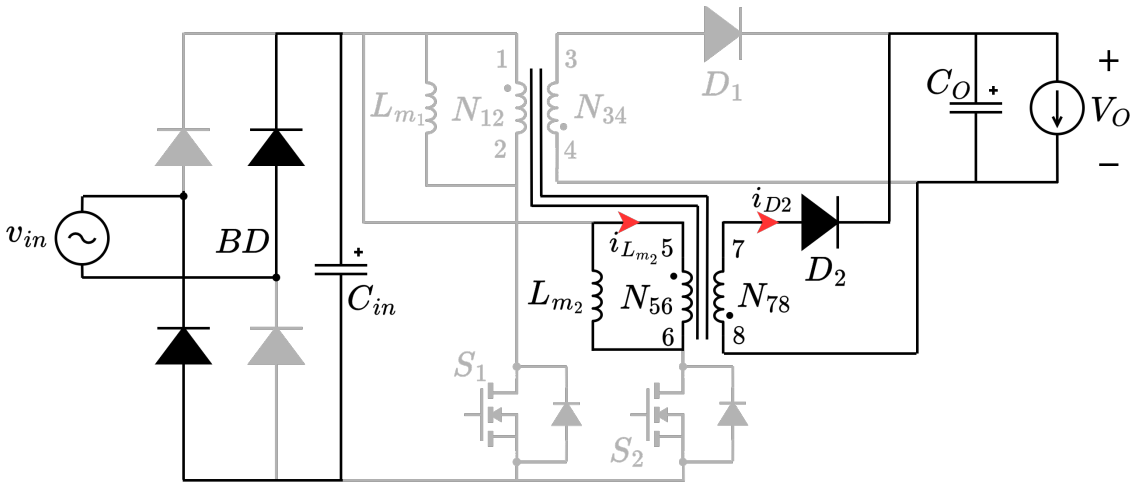


Figure 3.11: Operation of the proposed circuit with the transformer at Mode 6, $t_6 < t < t_7$

Mode 6, $t_6 < t < t_7$: S_2 turns off, signifying that $i_{L_{m_2}}$ has reached its peak value. The transformer $N_{56}N_{78}$ has finished charging and has started to discharge. i_{D_2}, $i_{L_{m_2}}$, and the flux densities B_l'', B_c'' and B_r'' start decreasing linearly. The corresponding currents and flux densities for transformer $N_{12}N_{34}$ stay at 0, since it is inactive during this time period and S_1 is still off. A representation of the active components in this mode are shown in Figure **??**.

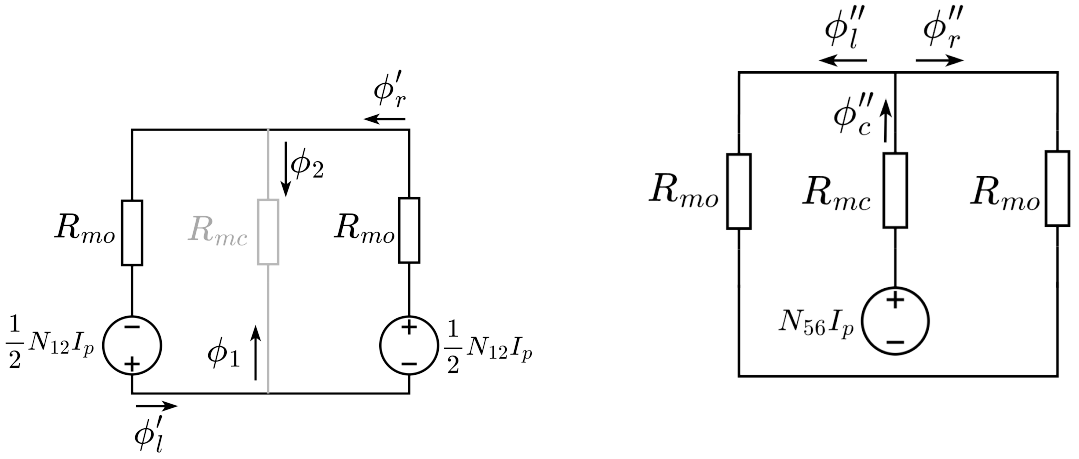
No coupling implicates that the operation of one transformer does not affect the other during the discharge stage. More specifically, the operation of one transformer does not affect the flux density of its partner transformer. This then implies the following:

1. The i_{Lm}, B waveforms, which are directly related to i_D should be linear in increase and decrease.

2. After discharge, the mentioned waveforms should stay at 0.

3. The inactive transformer should have $i_D = 0$.

Therefore, one could examine i_{D1} and i_{D2} explicitly to determine if the circuit operation includes coupling. This will be employed in the simulation and results verification section. This phenomenon is easily seen in modes 3 and 6.



(a) Outer Transformer $N_{12}N_{34}$. (b) Outer Transformer $N_{12}N_{34}$.

Figure 3.12: Equivalent Magnetic Circuits for the Proposed Transformer.

The operation of the transformer at modes 3 and 6 are further examined.

For Mode 3, the transformer being realized is shown in Figure **??** and the corresponding magnetic circuit is shown also in Figure **??**. For the positive cycle, the proposed winding causes the following: Since the magnetomotive force and magnetic resistance of both outer legs are equal, the magnetic flux flowing to the center

leg is canceled, resulting in no electromotive force being generated in transformer N_{56} and N_{78}. Consequently, the transformer formed by the outer legs can operate independently from the transformer in the center leg.

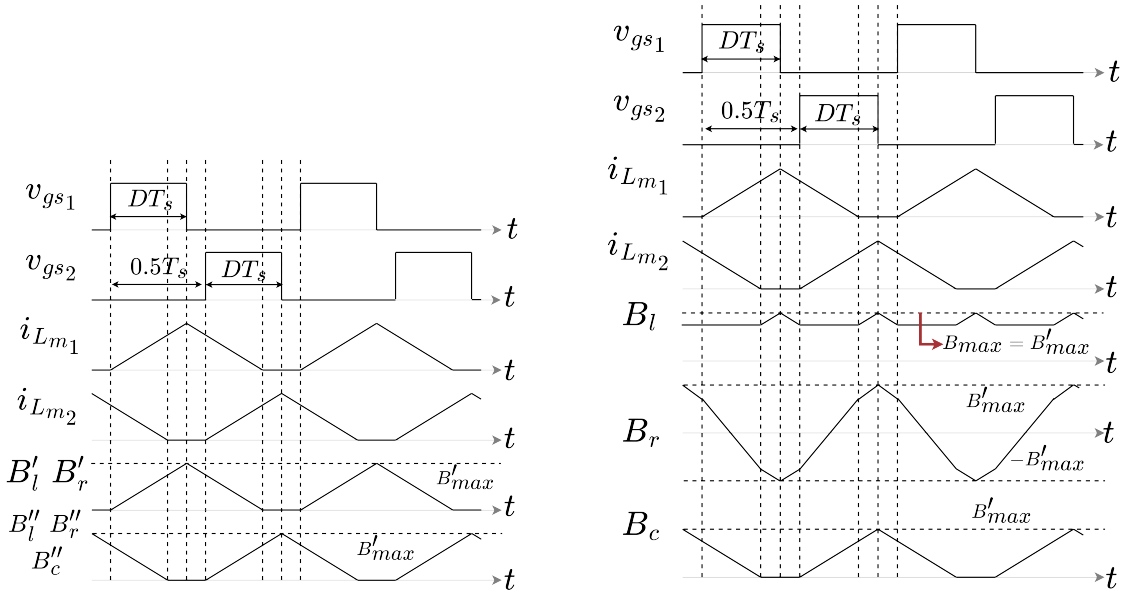
For Mode 6, the transformer being realized is shown in Figure **??** and the corresponding magnetic circuit is shown also in Figure **??**. For the negative cycle, the proposed winding causes the following: Since the magnetic resistance in the outer legs is equal, the flux flowing through the outer legs is also equal. Additionally, because the number of turns is equal and of opposite polarity in the outer legs, no net voltage is induced in the outer legs, making the other transformer non-operational. The center leg transformer can then operate separately from the outer leg transformer.

The proposed winding configuration results in a zero-coupling coefficient between the transformers, allowing them to operate separately in two phases.

3.2 Saturation Magnetic Flux Density Analysis

This section will detail the total magnetic flux density in the core in different operation patterns, as affected by the duty cycle. As mentioned in the previous section, the principle of superposition can be used to obtain the total.

The other two operational patterns will not be explained in detail, yet the superimposed flux density waveforms are shown in this section, along with the saturation magnetic flux densities of each case, which is important for the transformer design.



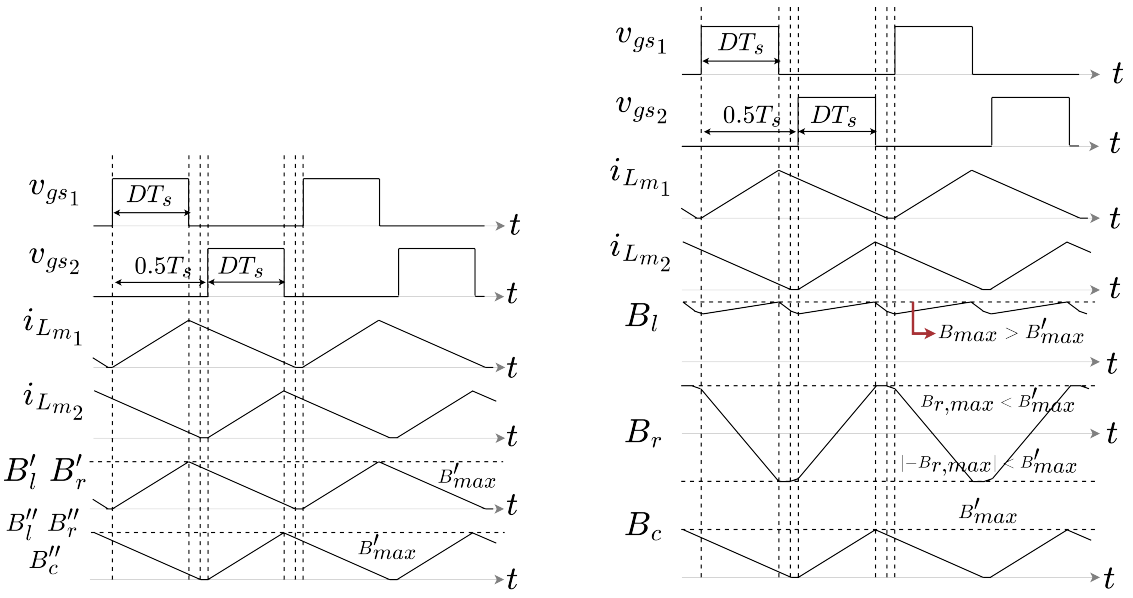
(a) Separated Waveforms. (b) Superimposed Waveforms.

Figure 3.13: Magnetic Flux Density Waveforms for Case 1.

For Case 1: $D \leq 0.5$ and $D \leq 0.5\frac{nV_O}{\sqrt{2}V_{\text{in, max}}}$. Here, one transformer discharges before the other fully charges, or its magnetizing inductance reaches the peak.

Superimposing the flux densities in Figure **??** with each other will then result in the following waveforms in Figure **??**. The left and right outer legs are not capable of magnetic flux cancellation. However, the peak fluxes can be seen in all legs. Therefore, the the saturation magnetic flux density of the core across all the phases is shown in Equation **??**.

$$B_{max} = B'_{max} \tag{3.2}$$



(a) Separated Waveforms. (b) Superimposed Waveforms.

Figure 3.14: Magnetic Flux Density Waveforms for Case 2.

For Case 2: $D \leq 0.5$ and $D \geq 0.5\frac{nV_O}{\sqrt{2}V_{\text{in, max}}}$. Here, one transformer does not discharge fully before the other fully charges, or its magnetizing current reaches the peak.

Superimposing the flux densities in Figure **??** with each other will then result in the following waveforms in Figure **??**. The left and right outer legs are not capable of magnetic flux cancellation. However, the peak flux density can only be seen in the left outer leg, at t_2 and t_5. The saturation magnetic flux density across all phases can be derived as follows and is shown in Equation **??**.

At t_2, B'_l is B'_{max}

At t_2, B''_l is also the B'_l at $t = t_5$:

$$B''_l(t_2) = B'_l(t_5) = B'_{max} - \frac{B'_{max}}{t_6 - t_2}(t_5 - t_2) \tag{3.3}$$

From Figure **??** , $t_5 = 0.5T_s + t_1 = 0.5T_s + DT_s$ and $t_2 = DT_s$.

To get t_6, one can examine the increase and decrease of the magnetizing inductance current during charge and discharge of the transformer. The peak, $I_{L_{m_1,\,\text{pk}}}$,

can be rewritten in terms of the charging and discharging side:

$$I_{L_{m_1,\,\mathrm{pk}}} = \frac{V_{\mathrm{in,\,max}}}{L_m} DT_s = \frac{nV_O}{L_m}(t_6 - t_2) \tag{3.4}$$

Solving equation **??** for t_6 yields:

$$t_6 = \frac{V_{\mathrm{in,\,max}}}{nV_O} DT_s + t_1 \tag{3.5}$$

$$= \frac{V_{\mathrm{in,\,max}}}{nV_O} DT_s + DT_s \tag{3.6}$$

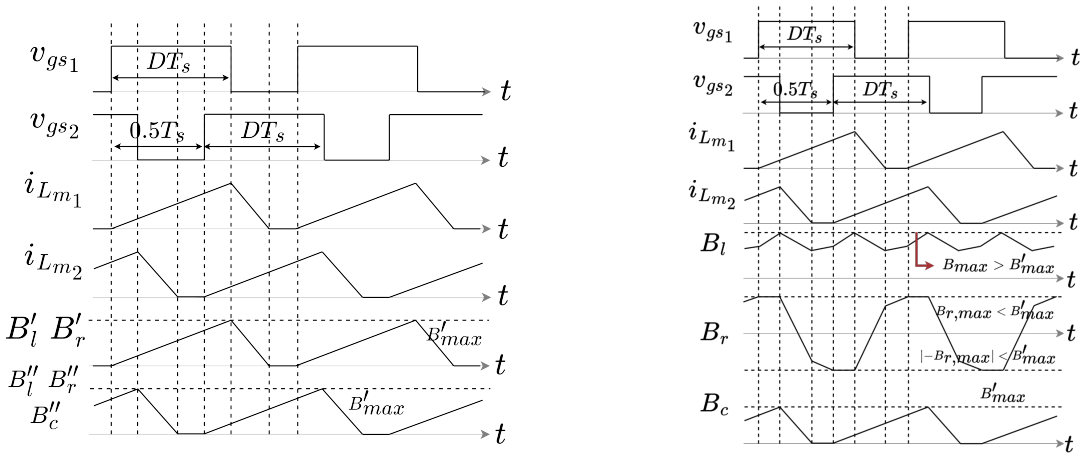
Using Equation **??** and **??**, the B_l'' at $t = t_2$ is given as:

$$B_l''(t_2) = B_l'(t_5) = B'_{max} - \frac{0.5B'_{max}nV_O}{V_{\mathrm{in,\,max}}D} \tag{3.7}$$

And therefore, the superimposed B_max found at the left outer leg is given by Equation **??**, at t_2 and t_5:

$$B_{max} = B''_{l_{max}} = \left(2 - \frac{0.5nV_O}{V_{\mathrm{in,\,max}}D}\right) B'_{max} \tag{3.8}$$

To realize the boundary condition between case 1 and 2, as mentioned, the transformers must discharge before reaching the peak magnetizing current.



(a) Separated Waveforms. (b) Superimposed Waveforms

Figure 3.15: Magnetic Resistances for the proposed transformer.

For Case 3: $D \geq 0.5$ and $D \geq 0.5\frac{nV_O}{\sqrt{2}V_{\text{in, max}}}$. Here, the transformers charge in succession and discharge before the next charge cycle of the other transformer.

Similar to Pattern 2, superimposing the fluxes in Fig. **??** with each other will then result in the following waveforms in Fig. **??**. The left and right outer legs are not capable of magnetic flux cancellation.The left and right outer legs are not capable of magnetic flux cancellation. However, the peak flux density can only be seen in the left outer leg, at t_2 and t_5. The saturation magnetic flux density across all phases can be derived as follows and is shown in Equation **??**.

At t_2, B'_l is also the B''_l at $t = t_5$::

$$B'_l(t_2) = B''_l(t_5) = \frac{B'_{max}}{DT_s}(t_5 - t_4) \tag{3.9}$$

At t_2, B''_l is B'_{max}. From Figure **??**, $t_5 = DT_s$ and $t_4 = 0.5T_s$.

Therefore, $B'_l(t_2)$ is as follows:

$$B'_l(t_2) = B''_l(t_5) = \frac{B'_{max}}{DT_s}(DT_s - 0.5T_s) \tag{3.10}$$

And therefore, the superimposed B_{max} found at the left outer leg is given by, at

t_2 and t_5:

$$B_{max} = B''_{l_{max}} = \left(1 + \frac{D - 0.5}{D}\right) B'_{max} \tag{3.11}$$

3.3 Miniaturization Effect of the Proposed Transformer

This section investigates the potential contribution of the proposed transformer to the volume reduction of the magnetic component. This section utilizes a part of the analysis method by **VolumeComparison**. This section will then detail too the volume and area products of the conventional and proposed transformer.

Firstly, the area-product of the conventional and proposed transformer is derived. The Area-Product (AP) Method is a core selection method wherein the transformer geometric dimensions are decided based on the circuit specifications by a relation between relevant quantities, such as current density and magnetizing inductance, making it a suitable way to determine the volume of the transformer. More specifically, the product of the window area A_w and magnetic path cross sectional area A_c, called the Area Product (AP) can be related to the circuit parameters. Choosing a core that has a bigger AP than the AP decided by the circuit parameters is ideal.

The analysis can be made simpler by making A_w and A_c a square shape **VolumeComparison**, which is shown here in Figure **??**. An EE-Core was chosen for this study. **For the EE-core transformer, it is assumed that the area of the center leg is twice that of the outer one**, corresponding to the industry standard.

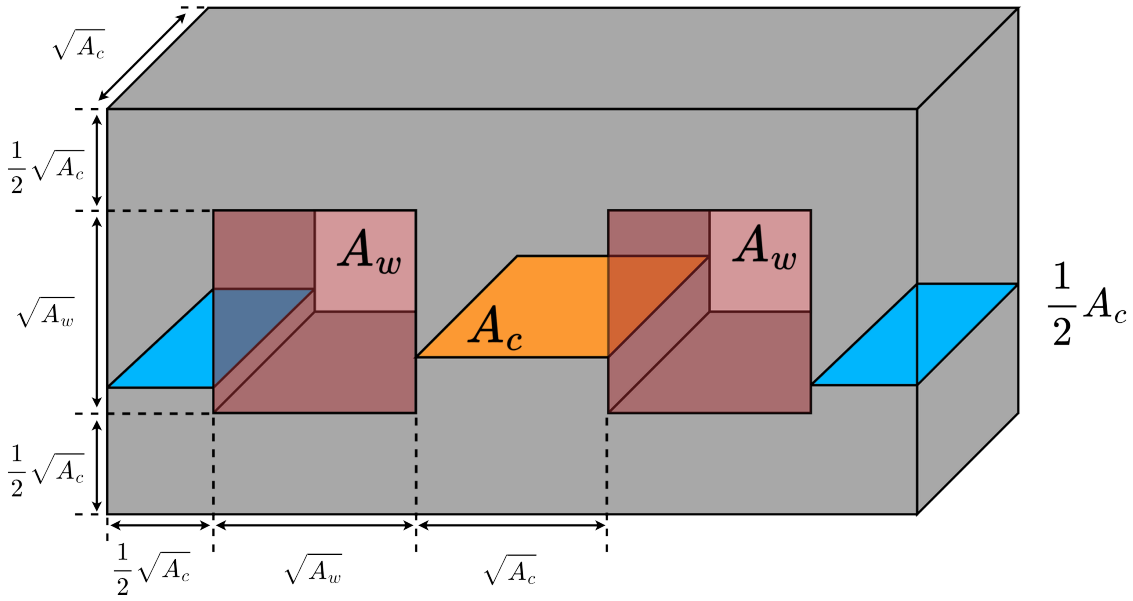


Figure 3.16: EE-core with a simplified A_w and A_c.

For the following analyses, the variables used are shown in Table **??**.

Variable	**Meaning**
L_m	Magnetizing Inductance
$I_{\text{p, pk}}$	Primary Side Peak Current
$I_{\text{p, rms}}$	Primary Side RMS Current
$I_{\text{s, rms}}$	Secondary Side RMS Current
n	Turns Ratio
J_m	Maximum Current Density
B_{max}	Maximum Flux Density
K_m	Occupancy Ratio

Table 3.1: Parameters used in the volume analysis.

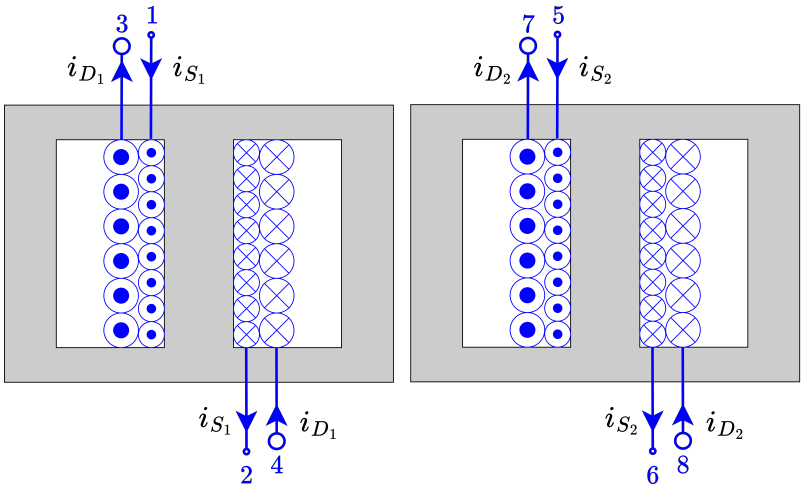


Figure 3.17: Representation of the conventional transformer with windings.

An analysis for the conventional transformer was first performed. The cross-section of the conventional transformer is shown in figure . One of the two transformers is analyzed first. Using Faraday's Law, an expression for A_c can be obtained as shown in Equation **??**:

$$\begin{aligned} N_{12} B_{max} A_c &= L_m I_{\text{p, pk}} \\ A_c &= \frac{L_m I_{\text{p, pk}}}{N_{12} B_{max}} \end{aligned} \tag{3.12}$$

On the other hand, the window area can be expressed in terms of the cross-sectional area of the windings shown in Equation **??**, with Faraday's Law, by taking it as a fraction of the window area A_w, denoted by the occupation factor K_u:

$$\begin{aligned} A_w K_u &= N_{12} \frac{I_{\text{p, rms}}}{J_m} + N_{56} \frac{I_{\text{s, rms}}}{J_m} \\ A_w K_u &= N_{12} \frac{I_{\text{p, rms}}}{J_m} + \frac{N_{12}}{n} \frac{I_{\text{s, rms}}}{J_m} \\ A_w &= \frac{N_{12}}{J_m K_u} \left(I_{\text{p, rms}} + \frac{I_{\text{s, rms}}}{n} \right) \end{aligned} \tag{3.13}$$

And therefore, the area product of one conventional transformer using Equation ?? and ?? is:

$$AP = A_c A_w = \frac{L_m I_{\text{p, pk}} \left(I_{\text{p, rms}} + \frac{I_{\text{s, rms}}}{n} \right)}{B_m J_m K_u} \tag{3.14}$$

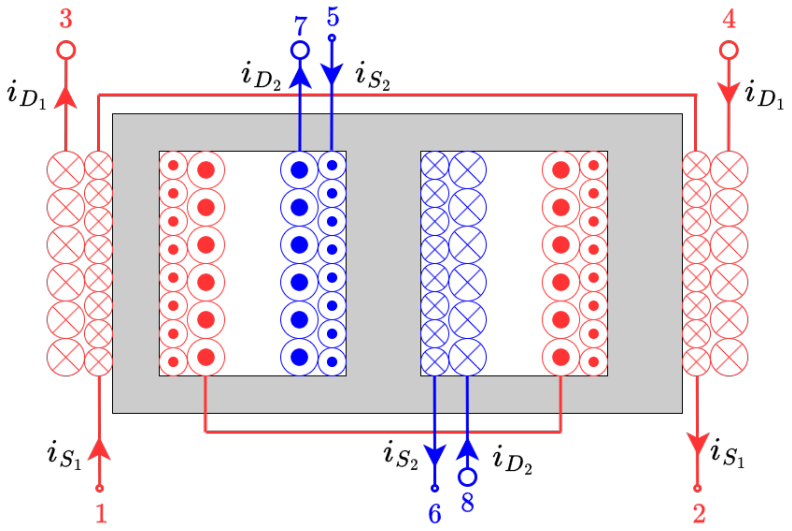


Figure 3.18: Representation of the proposed transformer with windings.

An analysis for the integrated transformer was performed next. The cross-section of the integrated transformer is shown in figure . Using Faraday's Law, an expression for A_c can be obtained in two ways as shown in Equation ??.

$$\begin{aligned} N_{12} B_{max} \frac{A_c}{2} &= L_m I_{\text{p, pk}} \\ N_{56} B_{max} A_c &= L_m I_{\text{p, pk}} \\ A_c &= \frac{2 L_m I_{\text{p, pk}}}{N_{12} B_{max}} = \frac{L_m I_{\text{p, pk}}}{N_{56} B_{max}} \end{aligned} \tag{3.15}$$

On the other hand, similarly, the cross-sectional area of the windings can be obtained using the following expression, with Faraday's Law. It is important to note that because of the assumption that the middle leg cross-sectional area is twice than that of the outer leg, then $N_{12} = 2N_{56}$ and $N_{34} = 2N_{78}$. Moreover, for the

proposed transformer, both half of the primary side winding and also the secondary side winding are occupying the window at the same time. This leads to the following expression in Equation **??**:

$$\begin{aligned} A_w K_u &= \left(N_{56} \frac{I_{\text{p, rms}}}{J_m} + N_{78} \frac{I_{\text{s, rms}}}{J_m} \right) + \frac{1}{2} \left(N_{12} \frac{I_{\text{p, rms}}}{J_m} + N_{34} \frac{I_{\text{s, rms}}}{J_m} \right) \\ A_w K_u &= N_{12} \frac{I_{\text{p, rms}}}{J_m} + \frac{N_{12}}{n} \frac{I_{\text{s, rms}}}{J_m} \\ A_w &= \frac{2N_{56}}{J_m K_u} \left(I_{\text{p, rms}} + \frac{I_{\text{s, rms}}}{n} \right) \end{aligned} \tag{3.16}$$

And therefore, the area product of the integrated transformer is:

$$AP = A_c A_w = \frac{\mathbf{2} L_m I_{\text{p, pk}} \left(I_{\text{p, rms}} + \frac{I_{\text{s, rms}}}{n} \right)}{B_m J_m K_u} \tag{3.17}$$

From Equation **??** and **??**, the area product of the integrated transformer is twice that of the conventional transformer.

$$AP_{integ.} = 2AP_{conv.} \tag{3.18}$$

The volume of the EE-core illustrated in Figure **??** can be written in terms of the area product AP and $x = \frac{A_c}{A_w}$:

$$\text{Volume} = 2AP^{\frac{3}{4}} \left(x^{\frac{3}{4}} + 2x^{\frac{1}{4}} + x^{-\frac{1}{4}} \right) \tag{3.19}$$

Finally, theoretically, the ratio of the volume of the integrated transformer and the proposed transformer can be calculated using equation **??**, **??**, **??**, and **??** as shown in equation 3.9. It is important to note that the conventional transformer is composed of two cores of the same kind, and thus its volume is twice than that of

one core.

$$\frac{\text{Volume}_{integ.}}{\text{Volume}_{conv.}} = \frac{2AP_{conv.}^{\frac{3}{4}}\left(x^{\frac{3}{4}} + 2x^{\frac{1}{4}} + x^{-\frac{1}{4}}\right)}{2AP_{integ.}^{\frac{3}{4}}\left(x^{\frac{3}{4}} + 2x^{\frac{1}{4}} + x^{-\frac{1}{4}}\right)} = 0.84 \tag{3.20}$$

This then shows that the proposed integrated transformer provides a 16% volume reduction compared to the conventional transformer.

Moreover, the expressions for $I_{\text{p, rms}}$ and $I_{\text{s, rms}}$ and their peaks are shown in Equation **??**. These can be used, as the circuit used in this study works similarly to that of the circuit in [].

$$I_{\text{p, rms}} = \frac{\sqrt{2}V_{\text{in, min}}D}{L_m f_s}\sqrt{\frac{D^3}{6}} \tag{3.21}$$

$$I_{\text{p, rms}} = \frac{\sqrt{2}V_{\text{in, min}}D}{L_m f_s}\sqrt{\frac{5nD^3}{9\pi} \cdot \frac{\sqrt{2}V_{\text{in, min}}}{V_O}} \tag{3.22}$$

Chapter 4

Transformer Design Method

This section provides a design method for the proposed integrated transformer. A core is selected using the area product method as described in Chapter 3. The method for choosing a core will not be covered anymore in this chapter. Moreover, it is assumed that the magnetizing inductances for the two transformers are equal $L_{m1} = L_{m2} = L_m$.

4.1 Turns Ratio

Firstly, a **turns ratio** n between the primary and secondary side is first decided. To make it simple, a turns ratio $n = \frac{N_P}{N_S}$ is decided based on the maximum voltage specifications of the chosen switching device. Fundamentally, the switch should withstand the peak voltage due to the effect of noise, magnetic induction, and the input voltage itself too. The following value of n can be obtained from the inequality shown in Equation **??**.

$$\begin{aligned} V_{\text{DS, max}} &\geq nV_O + \sqrt{2}V_{\text{in, max}} + V_{\text{spike}} + V_{margin} \\ n &\leq \frac{nV_O - \sqrt{2}V_{\text{in, max}} - V_{\text{spike}} - V_{margin}}{V_O} \end{aligned} \tag{4.1}$$

Here, $V_{\text{DS, max}}$ represents the maximum voltage specification of the switching device, V_O represents the output voltage, V_{spike} and V_{margin} represent some leeway for surge voltage and noise that the device may withstand and are typically set to $100V$ in various design applications.s

4.2 Saturation Magnetic Flux Density B_{max}

As have been discussed in Chapter 3, a specific magnetic flux pattern occurs in the transformer depending on **the duty cycle of the switches**. The desired flux operating pattern is first decided then afterwards a duty cycle for the circuit can be decided. There are therefore, three cases for the saturation magnetic flux density that occurs in the transformer, as shown in Equations **??** to **??**.

Case 1: $D < 0.5$ with $D \leq 0.5\frac{nV_O}{V_{\text{in, max}}}$

$$B'_{max} = B_{max} \tag{4.2}$$

Case 2: $D < 0.5$ with $D > 0.5\frac{nV_O}{V_{\text{in, max}}}$

$$B'_{max} = \frac{B_{max}}{2 - \frac{0.5nV_O}{V_{\text{in, max}}}} \tag{4.3}$$

Case 3: $D > 0.5$

$$B'_{max} = \frac{B_{max}}{1 + \frac{D-0.5}{D}} \tag{4.4}$$

We have to note that these cases are calculated in terms of the **peak of the maximum input voltage**. If the input voltage is less than the maximum, then one can have more leeway for the duty cycle to be used to operate in a chosen pattern.

4.3 Magnetizing Inductance L_m

The magnetizing inductance is decided using Faraday's law. The Magnetizing Inductance is an important parameter related to the amount of energy that is stored in the primary side of the transformer before being released to the secondary side.

In this study, the magnetizing inductance is set to be the same for the two sets of transformers. The magnetizing inductance is then obtained from Faraday's Law using the peak primary side current. Assuming that the output power is limited by the efficiency of the converter, then $P_O = \eta P_{in}$, with η being the efficiency of the converter, and P_{in} being the input power to the converter. We can then obtain the peak primary side current $I_{\text{pri, pk}}$ as shown in Equation **??**, using the output power P_O, efficiency η, and the **minimum** input voltage $V_{\text{in, min}}$.

$$I_{\text{pri, pk}} = \frac{0.5 P_O \sqrt{2}}{\eta V_{\text{in, min}} D_{max}} \tag{4.5}$$

To assume and make sure that the converter operates in DCM, in the Faraday's Law calculation shown below, the current is set to go from 0 to the peak primary side current in one cycle. Therefore, the required magnetizing inductance is then shown in Equation **??**. It is important to note that current rises and falls linearly in an inductor. Also, here, T_s represents the switching period of the converter while f_s represents the switching frequency.

$$\begin{aligned} V_{\text{in, min}} &= L_m \frac{\Delta i_{L_m}}{\Delta t} \\ &= L_m \frac{I_{\text{pri, pk}}}{D T_s} \\ L_m &= \frac{\sqrt{2} V_{\text{in, min}} D}{I_{\text{pri, pk}} f_s} \end{aligned} \tag{4.6}$$

4.4 Number of Turns N_{12}, N_{34}, N_{56}, N_{78}

After deciding the magnetizing inductance, the number of turns for each transformer can be decided using the relation between flux and self inductance as shown in equations **??** to **??**. The number of turns for the primary side of the transformers can be obtained using the turns ratio n. A geometric reference for the transformer is shown also in Figure **??**. The cross-sectional areas can be obtained using the dimensions of the core chosen.

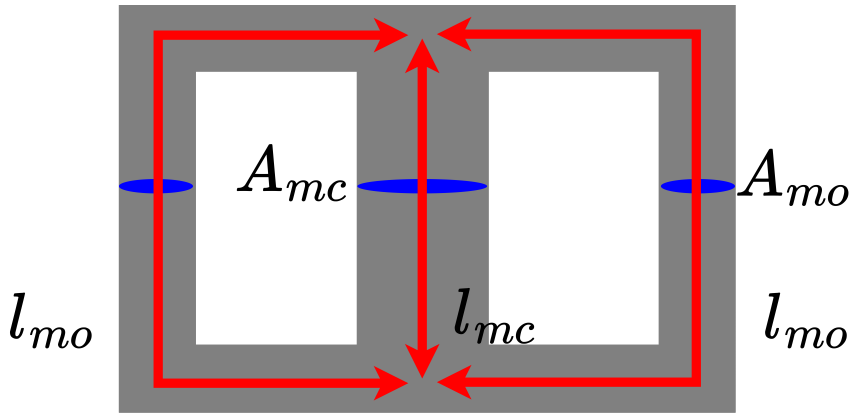


Figure 4.1: Geometric representation of the transformer.

For transformer N_{12}, N_{34} (outer leg transformer):

$$nN_{34}\Phi_{34} \geq L_m I_{\text{p, pk}}$$

$$N_{34} \geq \frac{L_m I_{\text{p, pk}}}{A_{mo} B'_{max}} \tag{4.7}$$

$$N_{12} = nN_{34} \tag{4.8}$$

For transformer N_{56}, N_{78} (center leg transformer):

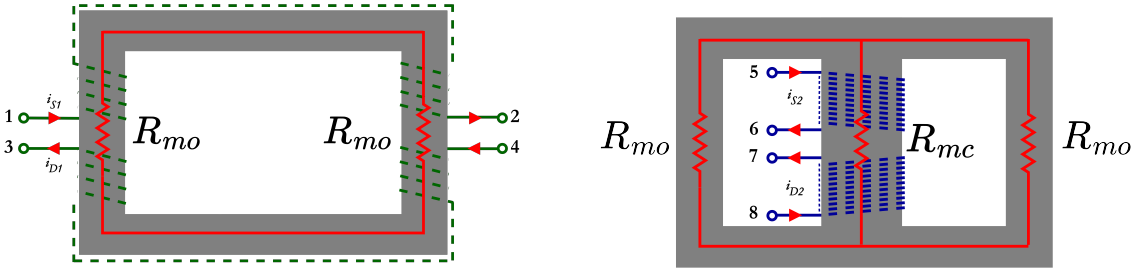
$$nN_{78}\Phi_{78} \geq L_m I_{\text{p, pk}}$$

$$N_{78} \geq \frac{L_m I_{\text{p, pk}}}{A_{mo} B'_{max}} \tag{4.9}$$

$$N_{56} = nN_{78} \tag{4.10}$$

4.5 Air Gap Design

An air gap is recommended to be inserted to avoid saturation of the core at any time. The air gap is going to be calculated based on the magnetic resistances. A representation of the magnetic resistances during the separate operation of the transformers is shown in Figure **??**.



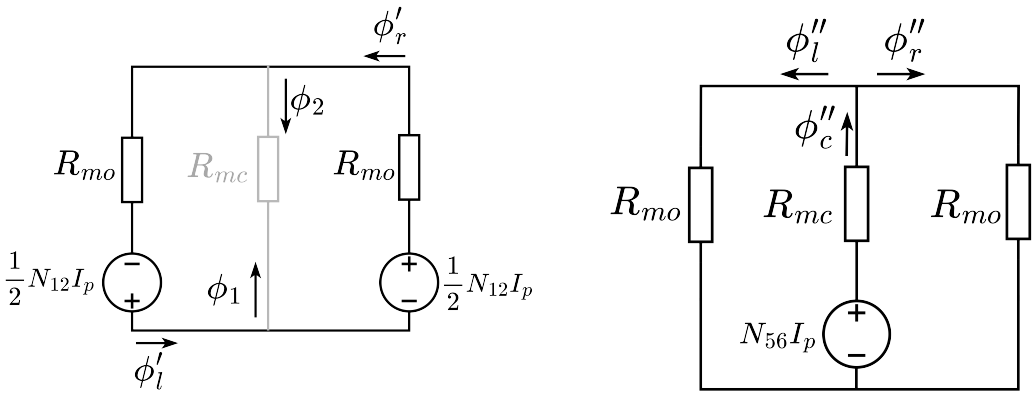
(a) Outer Transformer. (b) Center Transformer.

Figure 4.2: Magnetic Resistances for the proposed transformer.

For the outer leg transformer, as have been discussed in Chapter xx, the flux in the middle leg is canceled, and thus the equivalent circuit is shown in **??**, where it seems that there is no flux source in the middle leg. Moreover, since the magnetic resistances on the outer legs are equal, then the flux entering into the middle leg cancel each other out equally. Corresponding to the winding structure, the flux sources on the outer legs contain half the total number of turns wound on opposite polarity.

On the other hand, for the center transformer, as have been discussed, there is no net voltage produced on the outer legs, thus no flux sources are present there. Moreover, since the magnetic resistances on the outer legs are equal, then the flux entering the two outer legs should be equal to each other. The equivalent magnetic circuit is shown in figure **??**

From the magnetic circuits in Figure **??**, the expressions for the magnetic resistances can be derived.



(a) Outer Transformer. (b) Center Transformer.

Figure 4.3: Equivalent Magnetic Circuits for the Proposed Transformer.

For the outer leg magnetic resistance, the following expression for magnetizing inductance can be derived from figure **??**:

$$L_m = \frac{N_{12}^2}{2R_{mo}} \tag{4.11}$$

For the central leg magnetic resistance, the following expression for magnetizing inductance can be derived from figure **??**:

$$L_m = \frac{N_{56}^2}{R_{mc} + 0.5R_{mo}} \tag{4.12}$$

The expressions for R_{mo} and R_{mc} can be derived using **??** and **??**:

$$R_{mo} = \frac{N_{12}^2}{2L_m} \tag{4.13}$$

$$R_{mc} = \frac{(2N_{56} - N_{12}) + (2N_{56} + N_{12})}{4L_m} \tag{4.14}$$

The air gap can then be calculated. An expression for the magnetic resistances can be calculated using the air gap length and magnetic path length. However, two cases can be considered for the air gaps in the outer and center legs, denoted as $l_{mo_{air}}$ and $l_{mc_{air}}$, respectively, and are described below. This is because of the

non-uniformity of the flux path in the core between the central and outer legs.

Firstly, it can happen that the air gap $l_{mo_{air}}$ in the outer legs is bigger than that of the central leg $l_{mc_{air}}$. The geometric representation of this case is shown in Figure **??**. The resulting expressions for the outer and central leg resistance R_{mo} and R_{mc} in terms of the air gap length are shown in equation **??**. Using 4.13 - **??** will result into the appropriate air gaps for this case shown in **??**. It is important that physical manipulation of the existing core by shaving the outer legs is needed to add the additional air gap for the outer legs, which can cause a challenge in manufacturability at the moment.

$$\begin{cases} R_{mo} = \dfrac{l_{mo} - l_{mo_{air}} + l_{mc_{air}}}{\mu_r \mu_0 A_{mo}} + \dfrac{l_{mo_{air}}}{\mu_0 A_{mo}} \\ R_{mc} = \dfrac{l_{mc}}{\mu_r \mu_0 A_{mc}} + \dfrac{l_{mc_{air}}}{\mu_0 A_{mc}} \end{cases} \tag{4.15}$$

$$\begin{cases} l_{mo_{air}} = \dfrac{\mu_0 \left(\mu_r A_{mo} R_{mo} - A_{mc} R_{mc}\right) + \frac{l_{mc}}{mu_r} - l_{mo}}{mu_r - 1} \\ l_{mc_{air}} = \dfrac{\mu_0 A_{mc} R_{mc}}{\mu_r - 1} - \dfrac{l_{mc}}{\mu_r} \end{cases} \tag{4.16}$$

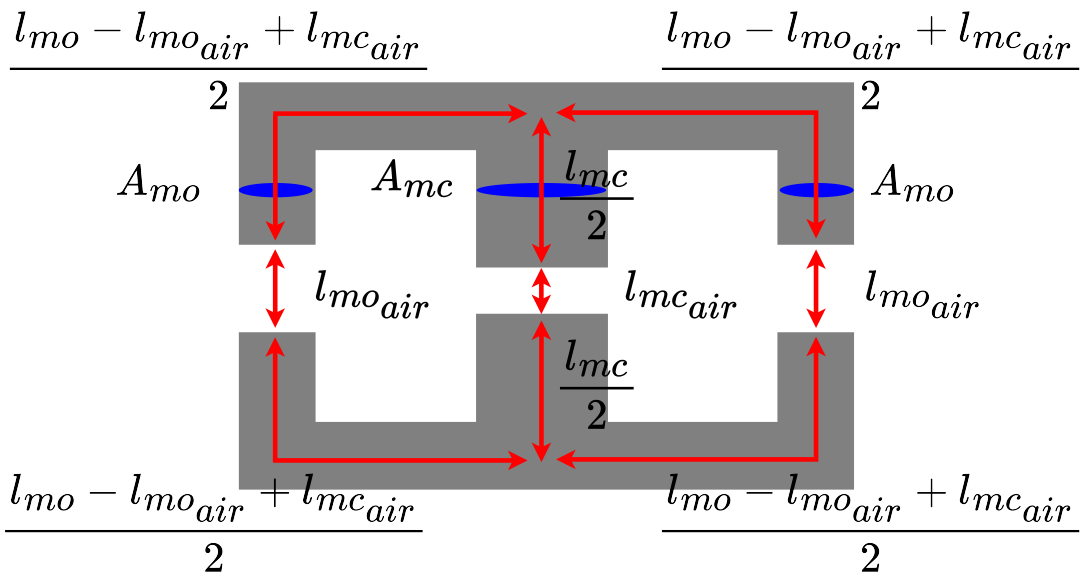


Figure 4.4: Proposed Transformer with $l_{mo_{air}} > l_{mc_{air}}$.

In the case that the air gap $l_{mc_{air}}$ in the central leg is bigger than that of the outer legs $l_{mo_{air}}$. The geometric representation of this case is shown in Figure **??**. Similarly, the resulting expressions for the outer and central leg resistance R_{mo} and R_{mc} in terms of the air gap length are shown in equation **??**, and consequently, the appropriate air gaps for this case is shown in **??**. Shaving of the center leg is needed to add the additional air gap for the center legs.

$$\begin{cases} R_{mo} = \dfrac{l_{mo}}{\mu_r \mu_0 A_{mo}} + \dfrac{l_{mo_{air}}}{\mu_0 A_{mo}} \\ R_{mc} = \dfrac{l_{mc} - l_{mc_{air}} + l_{mo_{air}}}{\mu_r \mu_0 A_{mc}} + \dfrac{l_{mc_{air}}}{\mu_0 A_{mc}} \end{cases} \tag{4.17}$$

$$\begin{cases} l_{mo_{air}} = \dfrac{\mu_0 A_{mo} R_{mo}}{\mu_r - 1} - \dfrac{l_{mo}}{\mu_r} \\ l_{mc_{air}} = \dfrac{\mu_0 \left(\mu_r A_{mc} R_{mc} - A_{mo} R_{mo}\right) + \frac{l_{mo}}{mu_r} - l_{mc}}{mu_r - 1} \end{cases} \tag{4.18}$$

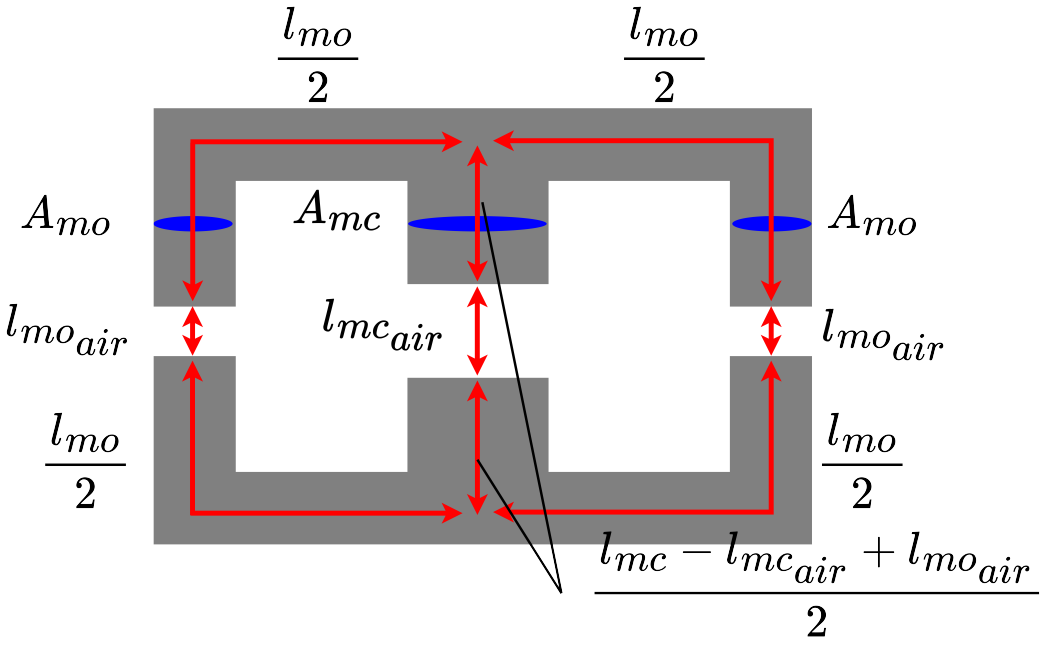


Figure 4.5: Proposed Transformer with $l_{mc_{air}} > l_{mo_{air}}$.

Chapter 5

Experimental Verification

This section describes experimental and simulation verification of the proposed transformer. Various wave forms and operational patterns will be shown in this section. For this study, the variation of the circuit operation **"Pattern 1"** was verified.

5.1 Prototype Circuit for Evaluation

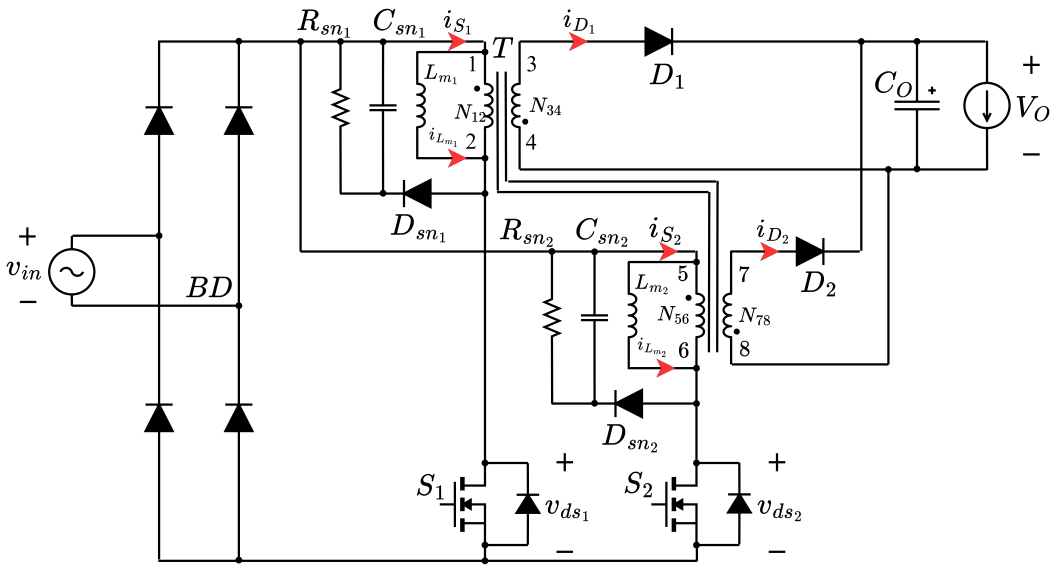


Figure 5.1: Prototype circuit for transformer design evaluation.

A prototype two-phase interleaved flyback rectifier, shown in **??**, was built to

verify the integrated transformer.

The circuit was operated in discontinuous current mode, where the phases with 180° difference. R_{sn_1} and R_{sn2} are snubber resistors, C_{sn1} and C_{sn2} are snubber capacitors, and D_{sn1} and D_{sn2} are snubber diodes, which form a clamped RCD snubber circuit [14]-[15]. This was added to protect the MOSFETs from surge voltages caused by transformer leakage inductance during turn-off of S_1 and S_2. Table **??** shows the circuit specifications and components used. The values for the snubber circuits are also fine-tuned using simulation. **The prototype circuit has no control stage for the output voltage yet since this study only focuses on a proof of concept first.**

Parameter	**Symbol**	**Value**
AC input voltage	v_{in}	100 V Vrms
Maximum Rated Power	$P_{O,\,\max}$	240 W
Output voltage	V_O	48 V
Switching frequency	f_s	100 kHz
Line frequency	f_s	60 Hz
Voltage Ripple	ΔV	8 V
Target Efficiency	η	90%

Table 5.1: Operating Conditions for the Converter.

Shown in Table **??** are the other circuit components used to accommodate the operating conditions of the two-phase interleaved flyback PFC rectifier.

Component	**Symbol**	**Value**
Bridge diode	BD	CD-DF408S
Input capacitor	C_{in}	1 μF
Switch	S_1, S_2	STF15N80k5
Secondary diode	D_1, D_2	STPSC30H12CWL
Output capacitor	C_o	2 x 1 mF (parallel)
Snubber capacitor	C_{sn_1}, C_{sn_2}	2.2 nF
Snubber resistor	R_{sn_1}, R_{sn_2}	30 kΩ
Snubber diode	$Dsn1, Dsn2$	VS-E7MH0112-M3/I

Table 5.2: Components used in the converter.

To design the output capacitor, the following formula was used and it depends on the output voltage ripple.

$$C_{\mathrm{O}} \geq \frac{P_O}{2\pi f_{\mathrm{line}} \Delta V V_O} \tag{5.1}$$

Finally, shown in Figure **??** is the prototype used in the experiment.

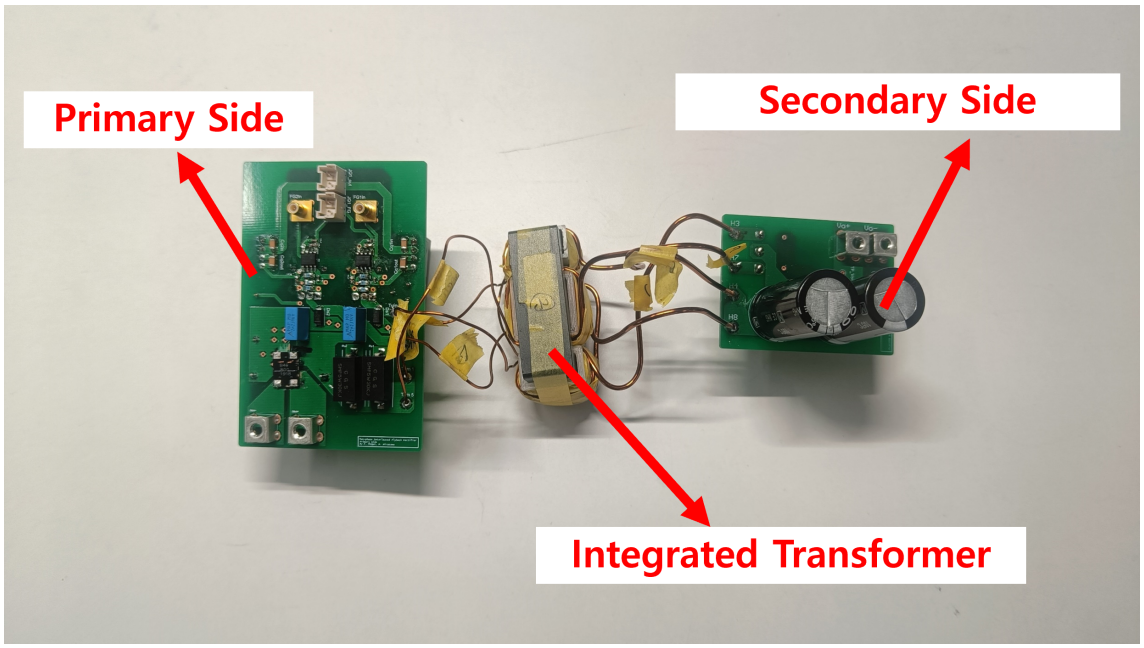


Figure 5.2: Actual prototype circuit for transformer design evaluation.

5.2 Prototype Transformer Design and Size Reduction

A prototype for the transformer was built. The specifications are shown in Table **??**.

The design process for this transformer is performed according to the method described in Chapter 4. The desired operating conditions for the converter are shown in Table **??**.

A switch with maximum V_{DS} at $800V$ was selected, with a margin and spike voltage of 100 V each. After using Equation **??**, a turns ratio n was selected to be 3. In the computation of **??**, a maximum input voltage of $V_{rms} = 240V$ was assumed for safety. Afterwards, with the operation pattern of **Pattern 1**, the maximum duty was assumed to be **0.47**, which will then provide the saturation magnetic flux density to be the same as that of the core specification. The magnetizing inductance required to support the **maximum** output load power is then **82.8** μH, obtained using Equation **??** and **??**, with the peak primary side current $I_{\text{pri, pk}}$ as 8.02 A.

A core bigger than the expected area product method was first chosen. Given what was available in the laboratory, the core 0P47721EC by Magnetics, inc. was selected as the core for this study. The area product of the core according to geometrical specifications is 46349 mm^2, and thus a suitable core was selected. The expected area product according to circuit specifications is shown in Table **??**. Using the equations described in Chapter 3, a 19% volume reduction is expected.

Item	Symbol	Conventional	Integrated
Magnetizing inductance	L_m	82.8 μH	
Turns ratio	n	3	
Primary side peak current	$I_{\mathrm{p,pk}}$	8.02 A	
Primary side rms current	$I_{\mathrm{p,rms}}$	2.25 A	
Secondary rms current	$I_{\mathrm{s,rms}}$	6.88 A	
Maximum flux density	B_{max}	0.3 T	
Maximum current density	J_{max}	5 A/mm^2	
Occupancy ratio	K_u	0.3	
Area product (circuit spec.)	AP	6703.21 mm^4	13406.42 mm^4
Core volume	Vol.	10313.57 mm^2	8672.64 mm^2

Table 5.3: Volume comparison using the actual core.

Finally then, a number of turns is selected. The number of turns were fine-tuned according to the inductance measured using the LCR Meter. The specifications of the transformer prototype are shown in Table **??**, and its diagram representation is shown in Figure **??**. The actual prototype manufactured is shown in Figure **??**.

Item	**Symbol**	**Value, Component**
Magnetic core	T	0P47721EC
Relative Permeability	μ_r	2500
Mag. path length (outer leg)	l_{mo}	71.4 mm
Mag. path length (center leg)	l_{mc}	31.7 mm
Cross-sectional Area (outer leg)	A_{mo}	117.6 mm^2
Cross-sectional Area (center leg)	A_{mc}	243.4 mm^2
Mag. inductance (Target)	L_m	82.8 μH
Mag. inductance of N_{12}, N_{34}	L_{m_1}	86.1086 μH
Mag. inductance of N_{56}, N_{78}	L_{m_2}	75.0598 μH
Turns ratio	n	3
Magnetic core	T	OP44721EC
Number of turns (outer primary)	N_{12}	24
Number of turns (outer secondary)	N_{34}	8
Number of turns (center primary)	N_{56}	15
Number of turns (center secondary)	N_{78}	5
Air gap (outer leg)	l_{go}	0.50 mm
Air gap (center leg)	l_{gc}	0.65 mm
Coupling coeff. of transformer N_{12}, N_{34}	$k_{N_{12}}, k_{N_{34}}$	0.9701
Coupling coeff. of transformer N_{56}, N_{78}	$k_{N_{56}}, k_{N_{78}}$	0.9787
Coupling coeff. between N_{12}, N_{34} and N_{56}, N_{78}	$k_{N_{12}}, k_{N_{56}}$	0.00053

Table 5.4: Specifications of the Prototype Transformer

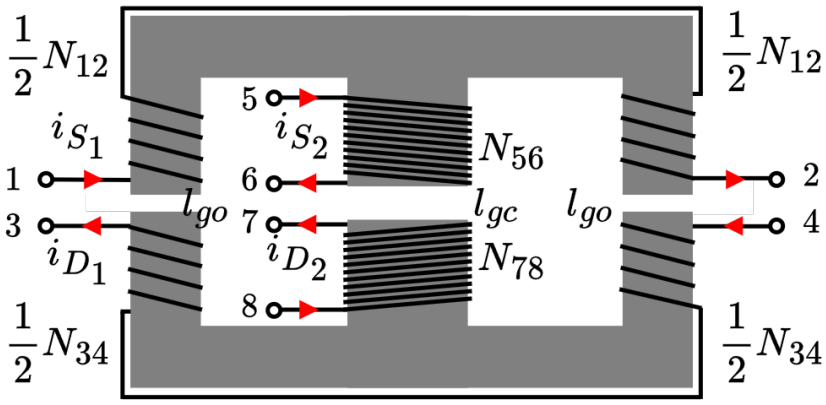


Figure 5.3: Diagram representation of the transformer.

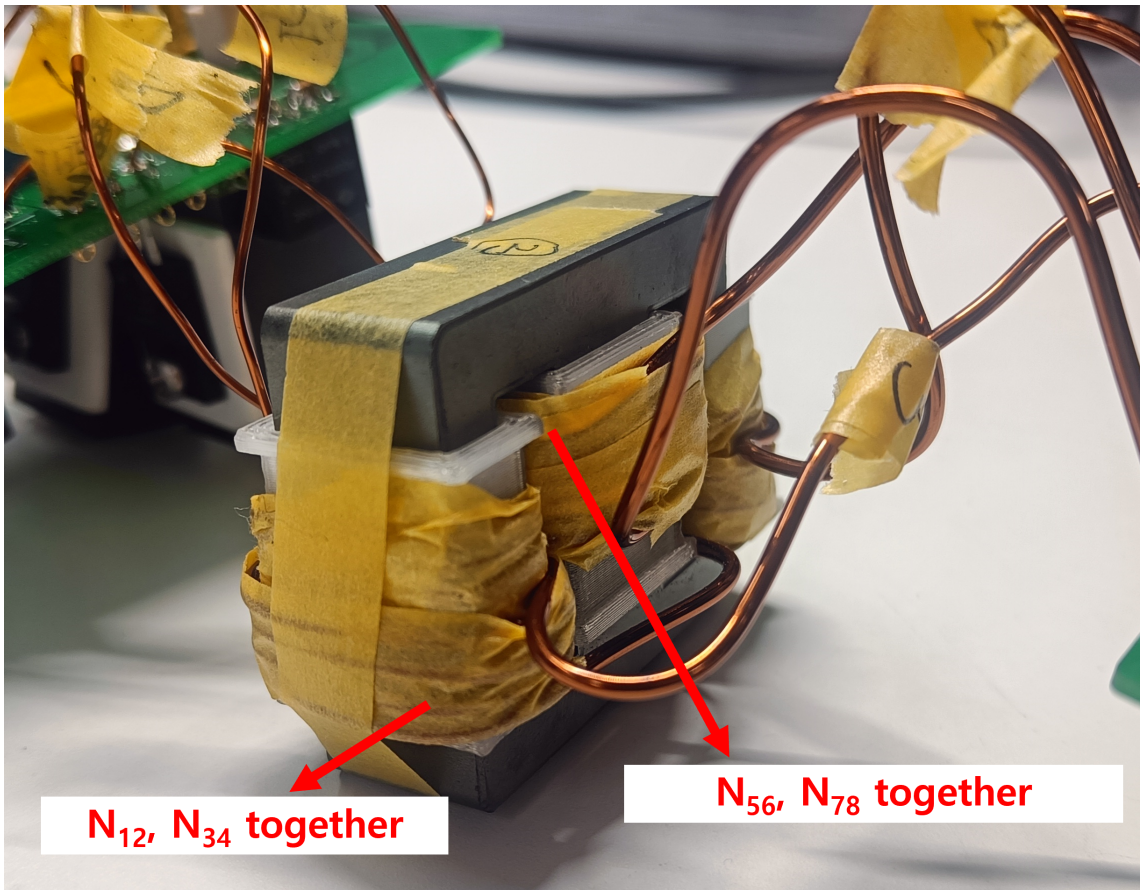


Figure 5.4: Actual prototype transformer.

The coupling coefficient measurement method is described as below. To take into account as much as possible the other parasitic inductances along the path of the inductors of the transformer, the following method is performed.

The involved inductors are first measured. Afterwards, they are connected in either series-aiding and series-opposing connection. This method aims to obtain the mutual inductance between the inductors.

For the series-aiding configuration measurement shown in Figure **??**, the total inductance measured is:

$$L_{meas\text{: series, aid}} = L_1 + L_2 + 2M \tag{5.2}$$

For the series-opposing configuration measurement shown in Figure **??**, the total inductance is measured as:

$$L_{meas\text{: series, oppose}} = L_1 + L_2 - 2M \tag{5.3}$$

From equation **??** and **??**, we can obtain the mutual inductance M as:

$$M = \frac{L_{meas\text{: series, aid}} + L_{meas\text{: series, oppose}}}{4} \tag{5.4}$$

And from here, one can obtain the coupling coefficient k between the two involved inductors as:

$$k = \frac{M}{\sqrt{L_1 L_2}} \tag{5.5}$$

And lastly, the magnetizing inductance of the transformer is obtained as:

$$L_m = k^2 L \tag{5.6}$$

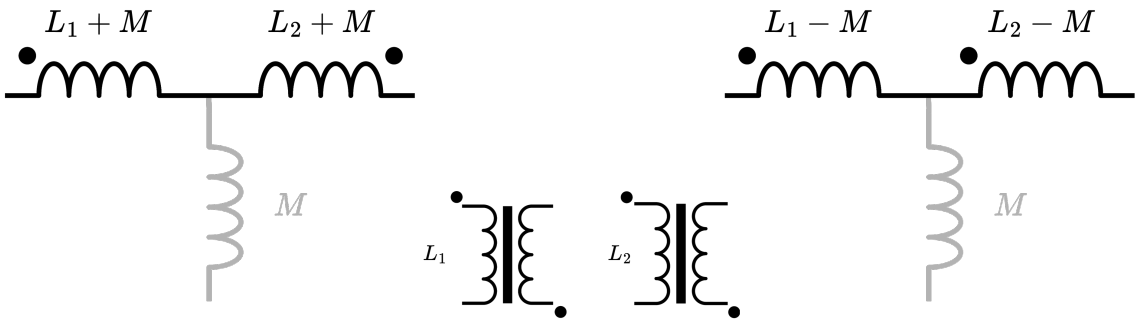


Figure 5.5: Inductance measurement setup.

5.3 Simulation Verification

Before creating the actual prototype, a verification through simulation was first performed. The prototype circuit shown first in **??** was made in PLECS. The resulting circuit is shown in Figure **??**.

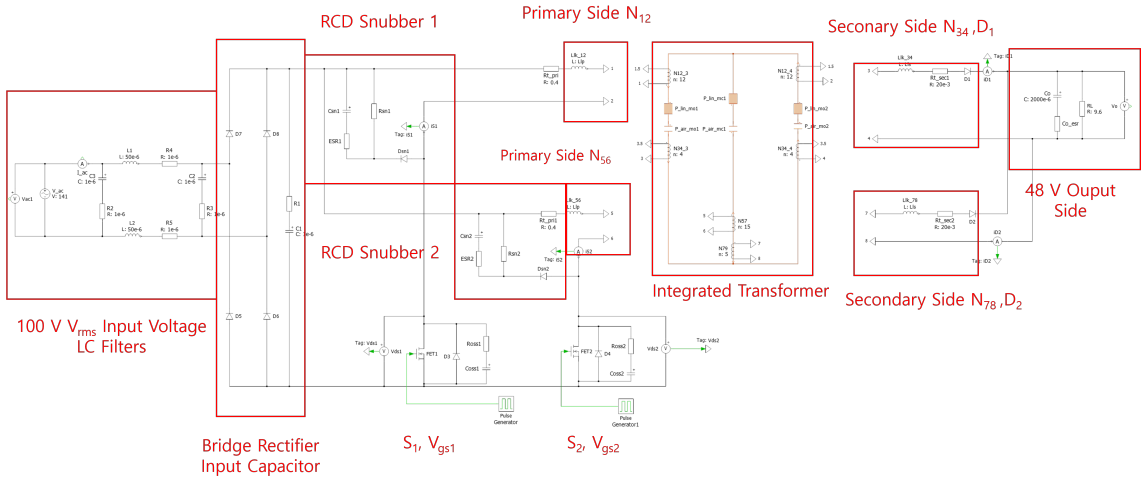


Figure 5.6: Simulation Circuit for the evaluation circuit in PLECS.

5.3.1 Flux Operation

The magnetic component designed in PLECS is shown in Figure **??**.

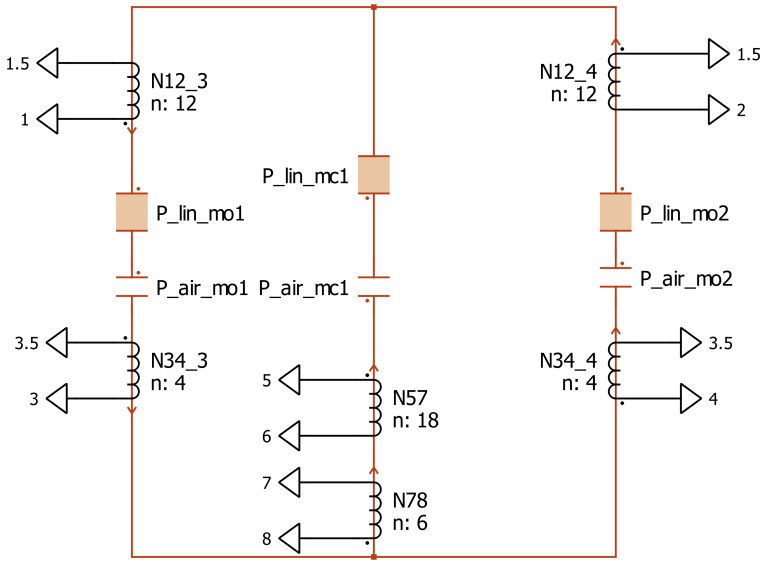
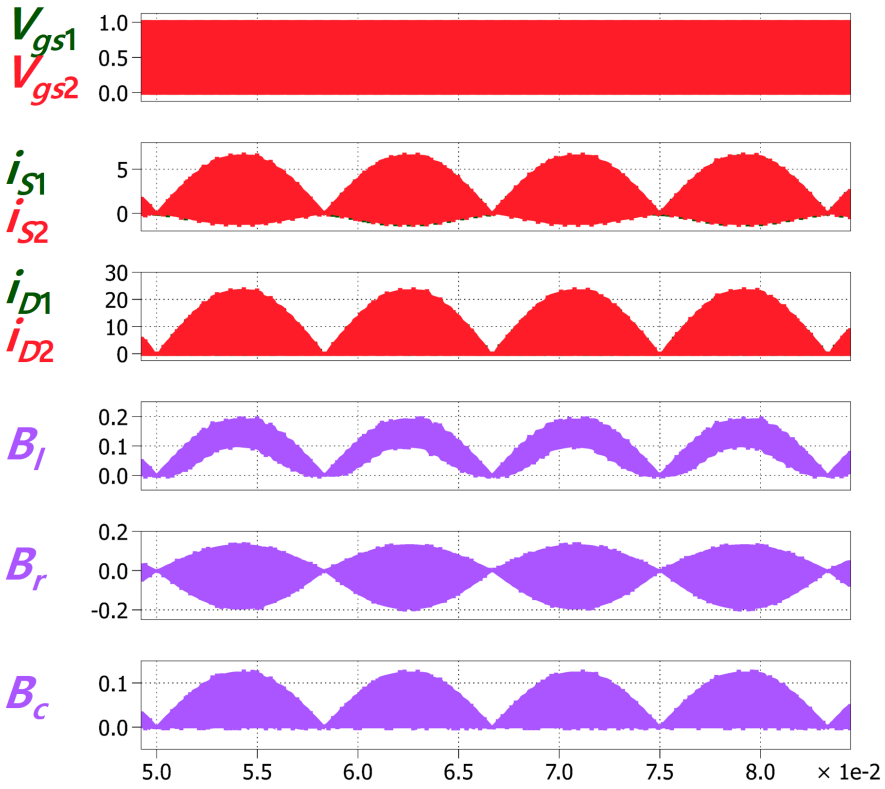
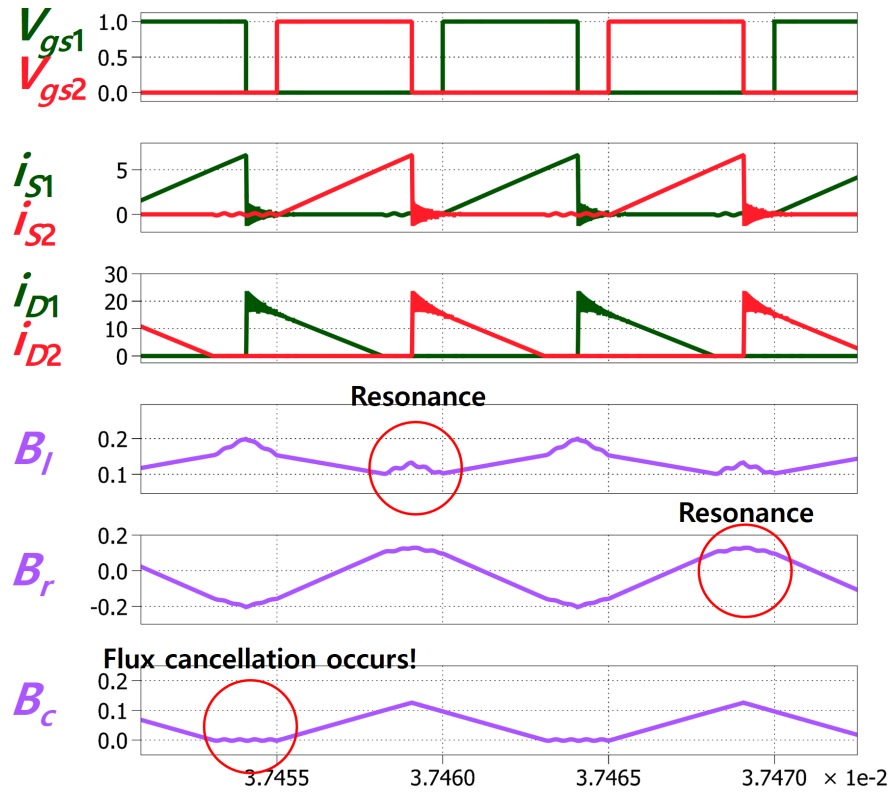


Figure 5.7: Proposed transformer in PLECS.

The waveforms obtained from the magnetic component designed is shown in Figure **??**, with a zoomed in version in Figure **??**. For Pattern 1, a somewhat similar shape can be observed to that of the theoretical analysis in Chapter 3 (see Figure**??**). Flux cancellation does not occur in any of the left and right outer legs, but flux cancellation is seen in the center leg. Peak flux is then seen in the left leg instead. The irregularities in the shape arise due to the unequal magnetic inductances present in the transformer. In the particular case shown in the figure, $L_{m_1} < L_{m_2}$ because L_{m_1} reaches a higher peak i_{S_1}. This causes the flux during the operation of N_{12}, N_{34} to rise and fall from a higher peak, which then causes the differences in the expected result theoretically. Although it was designed by simulation, the method used to create the magnetic element makes it hard to make the inductances equal.



(a) Zoomed-out flux density waveforms.



(b) Zoomed-in flux density waveforms.

Figure 5.8: Simulated flux density waveforms for the proposed transformer.

Nevertheless, flux cancellation occurs in the center leg, and also, the changes in magnetic flux are somewhat linear, except at the peaks, which are due to the resonance present between the leakage inductance of the transformers and the switch capacitances, which then implies that no coupling occurs between the two transformers and operate separately from each other during each phase.

To account for this non-ideal operation, the magnetic component has been changed to the component shown in Figure **??**, where the transformer is assumed to be one big coupled inductor composed of four different magnetic components. This time, it is ensured that the magnetizing inductances of the transformers are equal and will be used to simulate the remaining cases as will be described in the next subsections.

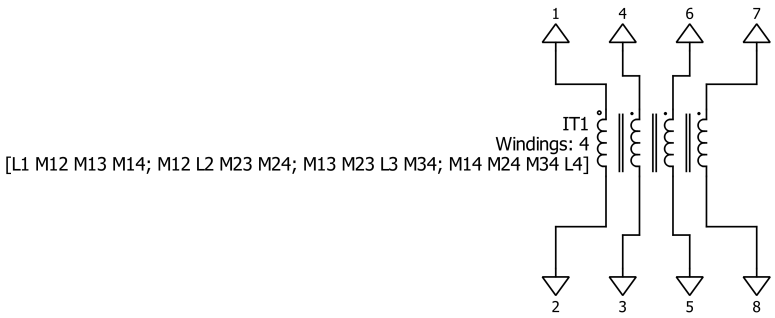
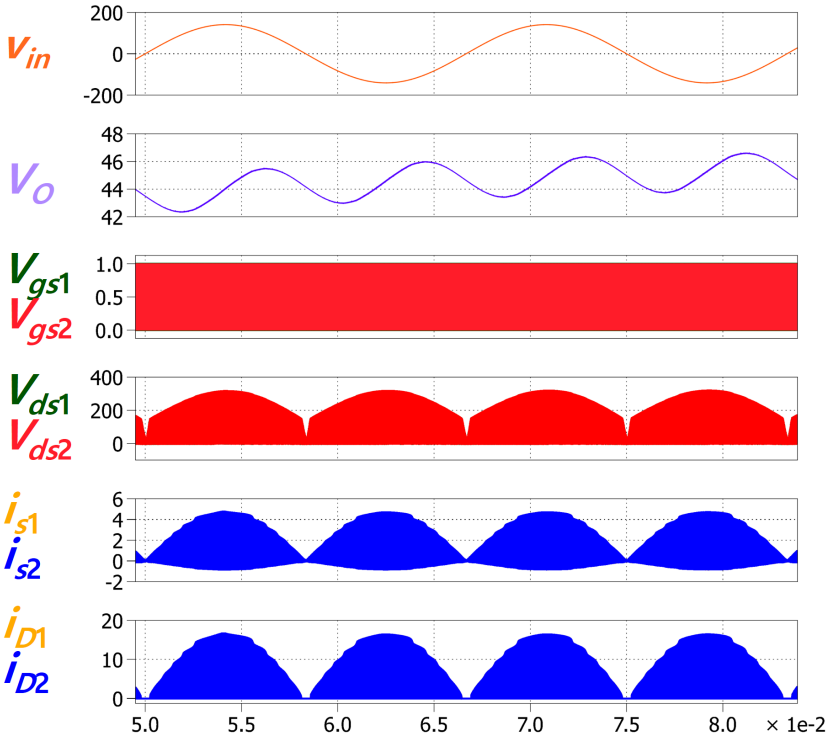


Figure 5.9: New magnetic element used in simulation.

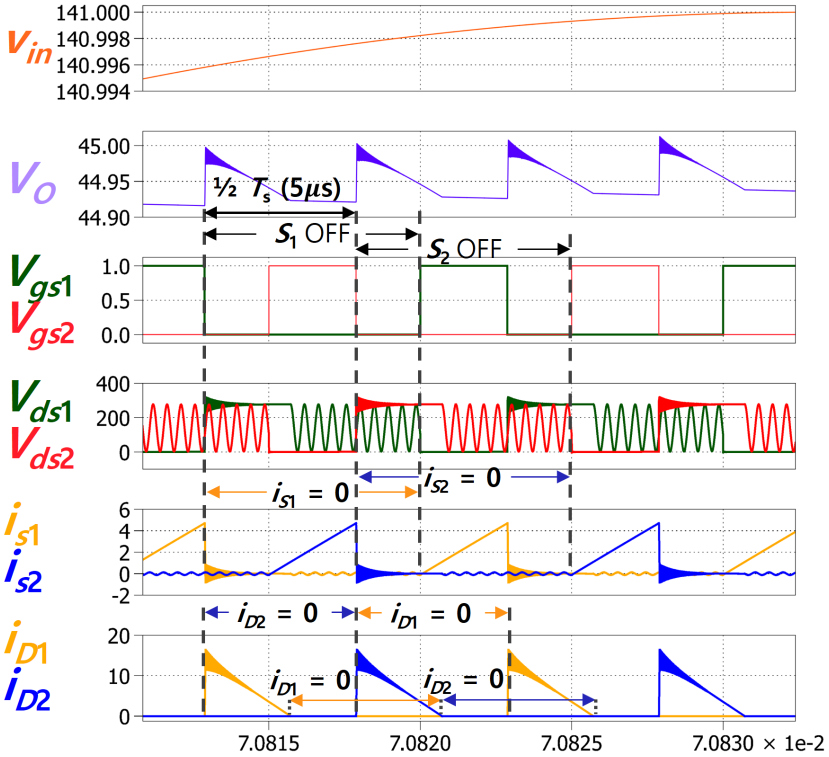
5.3.2 Two-phase Operation

Figure **??**shows waveforms from the simulation testing of the two-phase interleaved flyback rectifier with the proposed transformer at 100 W. Proper PFC Operation can be seen, especially that the current through the switches follow the same phase as of the rectified voltage. On the other hand, from the zoomed-in wave forms in Figure **??**, wherein during off-time of S_1 and S_2, the current i_{s_1} is zero during the off-time of S_1 and i_{s_2} is zero during the off-time of S2. No current is induced in either switch except during the resonance between the leakage inductance and the parasitic capacitance of the switch. During the inactive period of the transformer, for example, between the start of the off-time of S_1 and the start of the off-time of S_2, the secondary diode side current is zero. Moreover, after i_{D1} and i_{D2} reached zero when S_1 and S_2 switched off, the currents remained at zero. Lastly, a linear increase and decrease of i_{S_1}, i_{S_2}, i_{D_1}, and i_{D2} was observed throughout the circuit operation. Therefore, from simulation, the two flyback converters N_{12}, N_{34}, S_1, D_1, C_O, V_O and N_{56}, N_{78}, S_2, D_2, C_O, V_O can operate independently without magnetic coupling interference, as coupling was eliminated between the flyback transformers.

The same phenomenon was observed at low (10 W) and high loads (240 W), which therefore implies that the transformer can operate separately in two phases at any load with out any magnetic interference. The results are shown in Figures **??** and **??**.

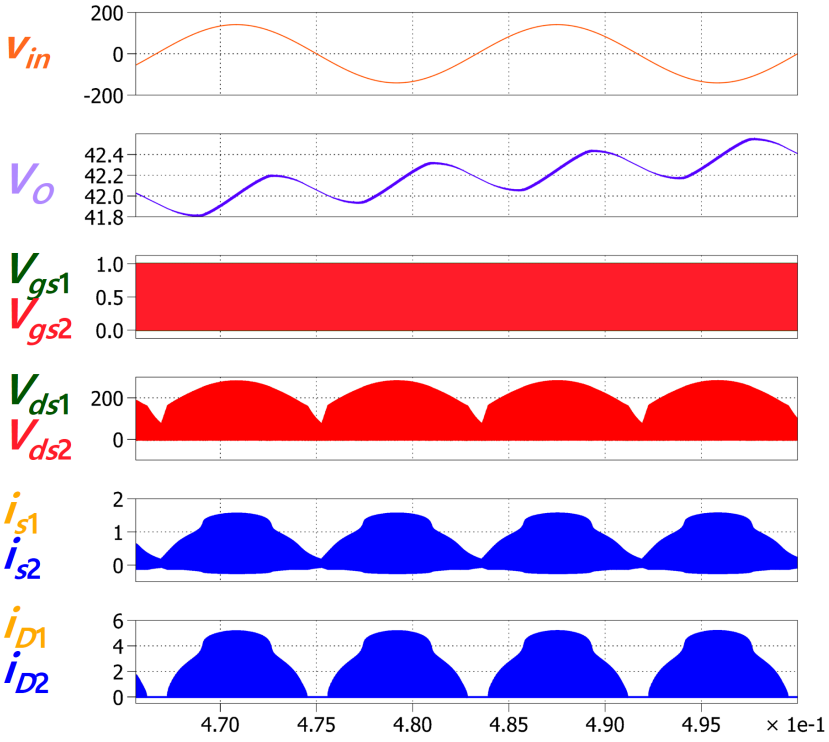


(a) Zoomed-out simulated waveforms.

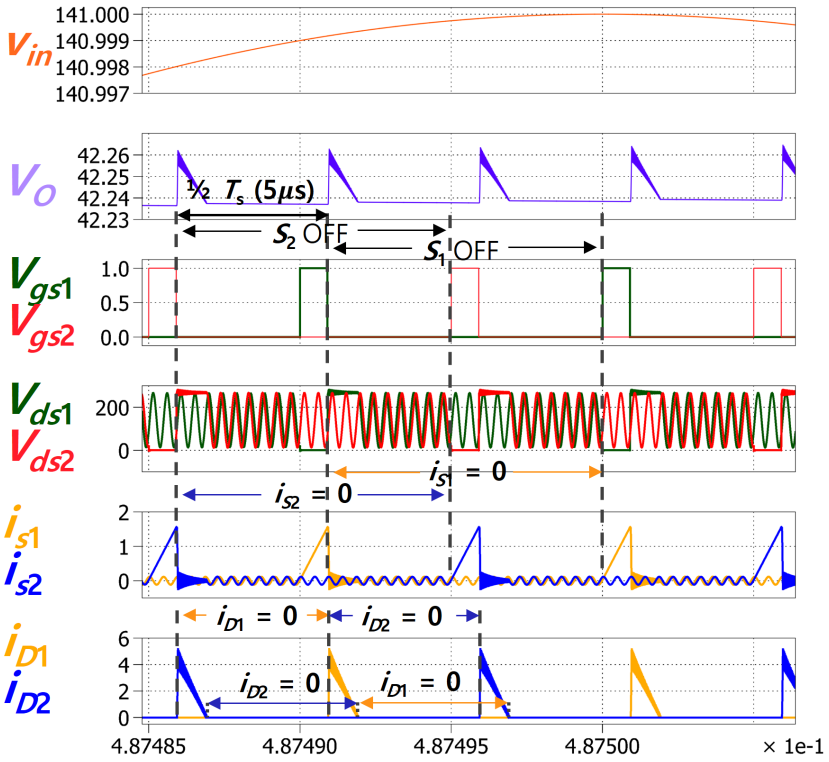


(b) Zoomed-out simulated waveforms.

Figure 5.10: Simulated waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at 100 W, two-phase AC drive.

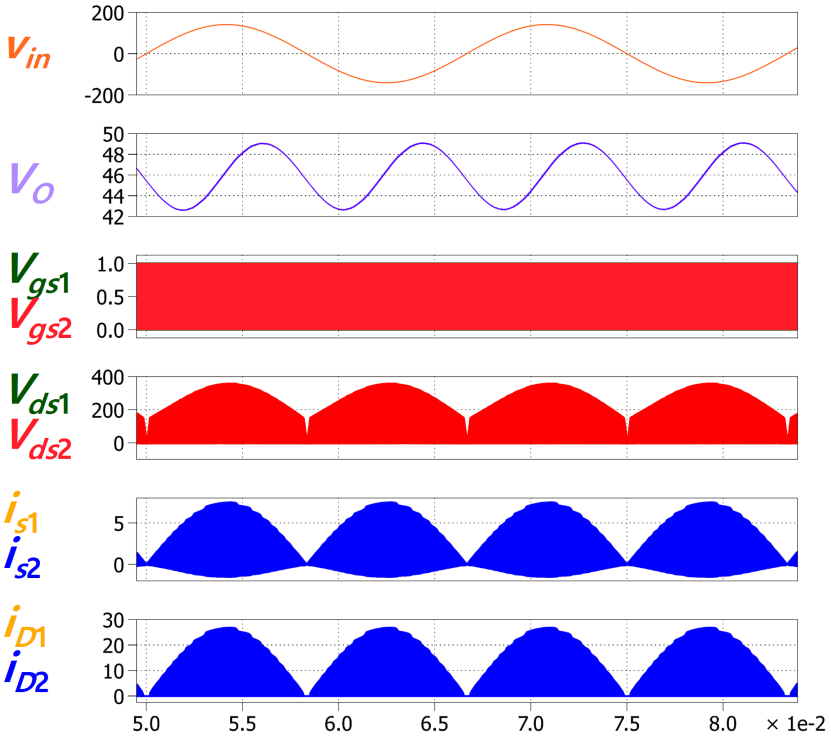


(a) Zoomed-out simulated waveforms.

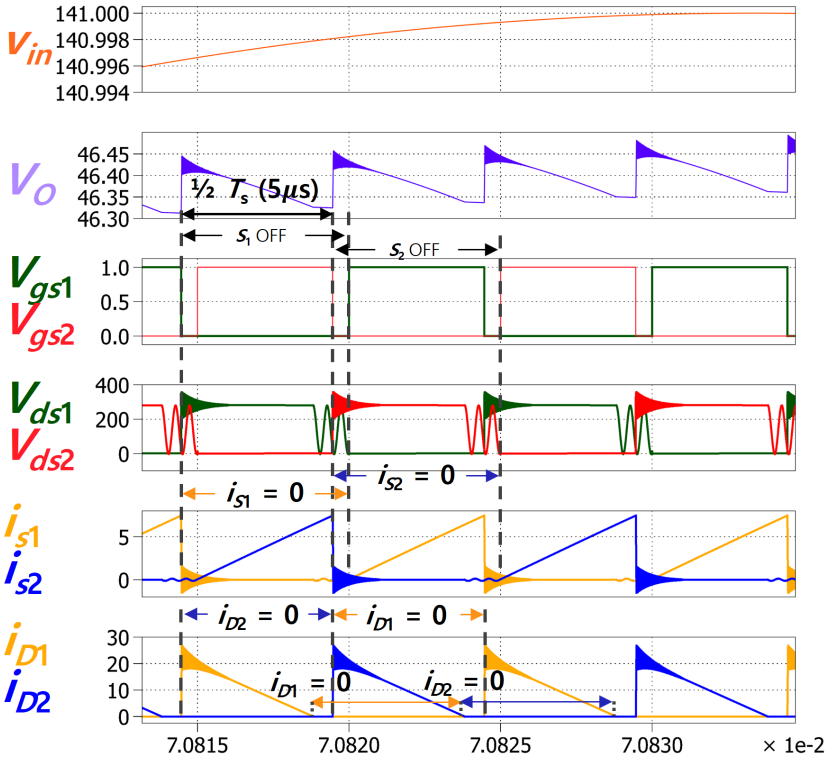


(b) Zoomed-in Simulated Waveforms.

Figure 5.11: Simulated waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at 10 W, two-phase AC drive.



(a) Zoomed-out simulated waveforms.

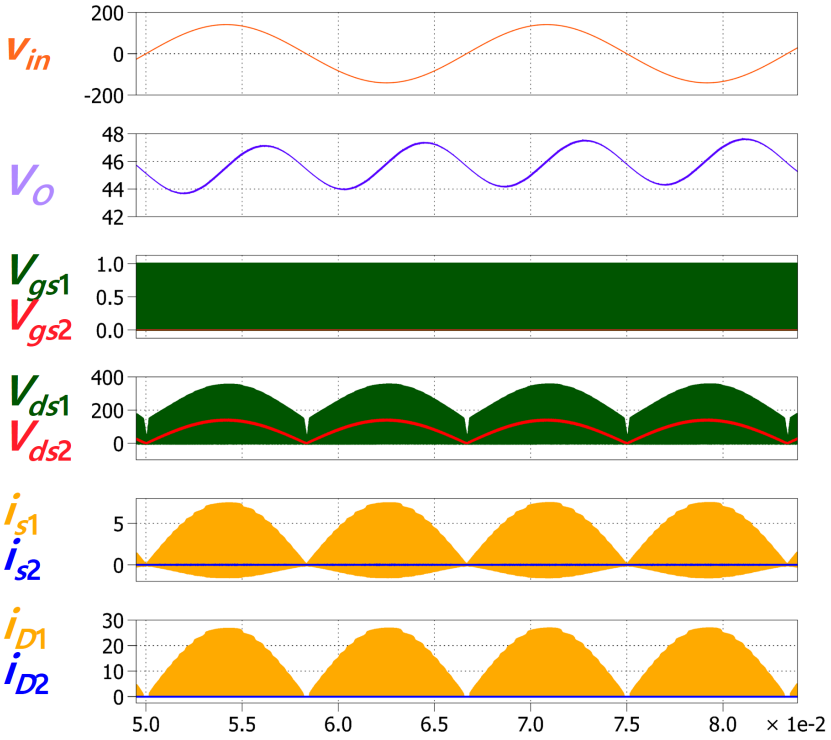


(b) Zoomed-in simulated waveforms.

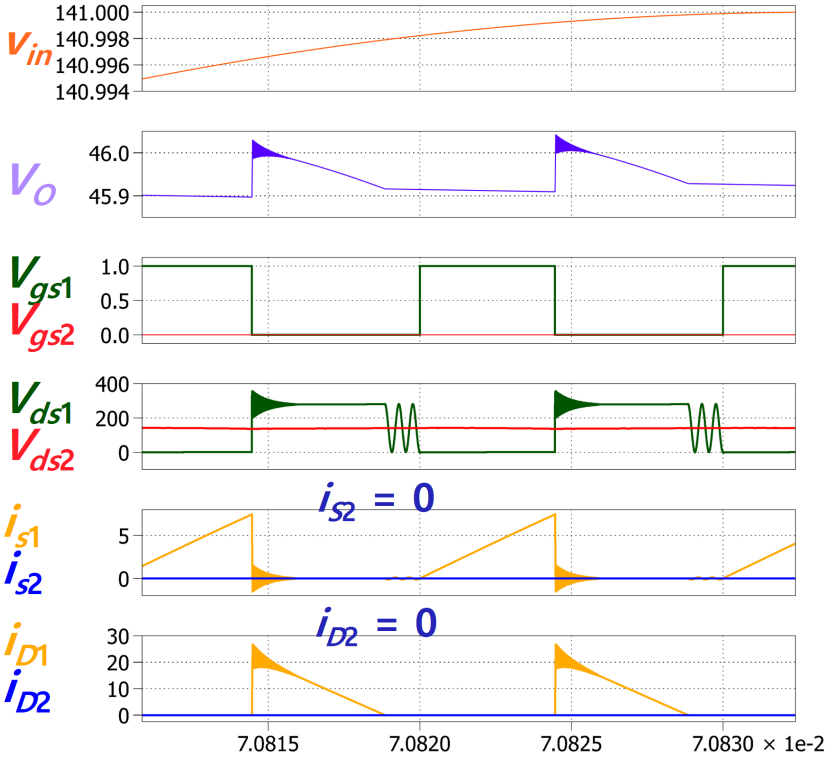
Figure 5.12: Simulated waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at 240 W, two-phase AC drive.

5.3.3 Single-phase Operation

Single-phase drive at lower power is also feasible with this transformer. Figure **??** shows waveforms from the simulation testing of the two-phase interleaved flyback rectifier with the proposed transformer at 120 W AC, with only driving S_1. PFC operation can still be observed from the zoomed-out view of the waveforms. Moreover, from the zoomed-in waveforms shown in **??**, i_{S_2} and i_{D_2} are zero all throughout the circuit operation. This then confirms that only the flyback converter N_{12}, N_{34}, S_1, D_1, C_O, and V_O are operating. There is a voltage seen at V_{DS_2} since the two flyback converters are in parallel, although the switch is kept off. Therefore, from simulation, non-influence of the inactive transformer to the active one is confirmed.



(a) Zoomed-out simulated waveforms.



(b) Zoomed-in simulated waveforms.

Figure 5.13: Simulated waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at 120 W, one-phase AC drive.

5.3.4 Comparison of Efficiency between the circuit with the Conventional and Proposed Transformer

To see if the integration of the transformer into one core has an effect on conversion efficiency, an efficiency analysis has been performed using PLECS across the power range 10 W - 240 W, at 100 V V_{rms}, 100 kHz switching frequency. Both the conventional and proposed simulated transformer are put into the circuit and then once at the steady state, the input and output power have been calculated, from where the efficiency can be obtained. The magnetic models used for the simulation are shown in Figure **??**.

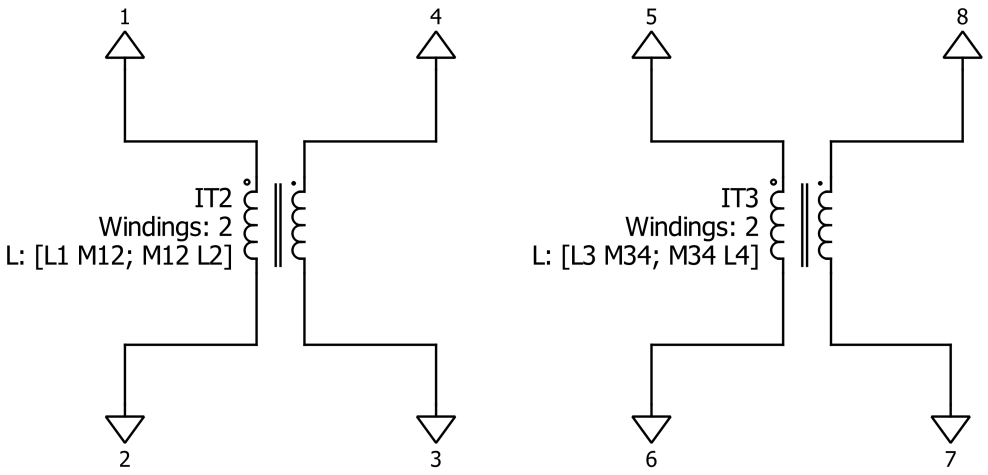
The results of the analysis are shown in Figure **??**. As seen, there is almost no difference between the efficiency of the converter when using the conventional and proposed transformer. Therefore, the integration into one core will not affect the efficiency too much, and it is better to use the integrated magnetic component instead, to allow for circuit miniaturization and reduction of construction cost.

However though, a detailed loss analysis of the core model has not been performed yet using Finite Element Analysis (FEM) methods and calculation using excitation. Especially that the operation of this transformer is special, then the losses that can be obtained from here may not be straightforward compared to that of traditional analysis methods. Therefore, this part is going to be investigated further and results will be used to optimize the design of the proposed transformer in this study.

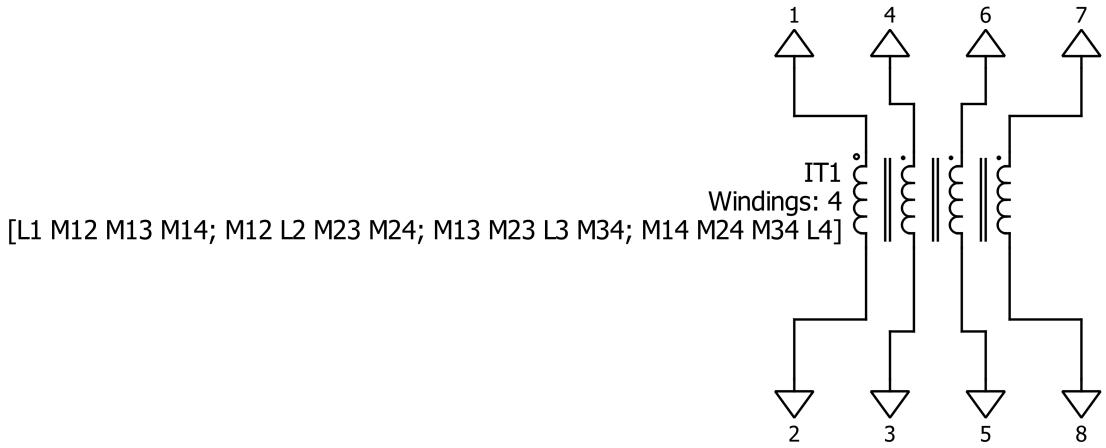
5.3.5 Investigation of the Effect of Coupling to Circuit Operation

The effect of coupling to the operation of the circuit has been observed through simulation. It is still difficult as of now to obtain the operational principle theoretically. Figure **??** shows waveforms from the simulation testing of the two-phase

interleaved flyback rectifier with the proposed transformer at 240 W, with coupling coefficient between the two transformers, also known as $k_{N_{12}N_{34},N_{56}N_{78}} = 0.8$. Proper PFC Operation can be seen, however, there are strange peaks seen throughout the waveform, especially at peak input voltage and regions near the lowest input voltage, implying a possible interaction between the operation in two phases. From the zoomed-in wave forms in Figure **??**,it is clearly seen that the transformers do not operate independently of each other in each phase. The decrease of i_D is not linear to zero, and overlaps a bit with the discharge of the other transformer, as labeled in the figure. Therefore, the two flyback converters N_{12}, N_{34}, S_1, D_1, C_O, V_O and N_{56}, N_{78}, S_2, D_2, C_O, V_O cannot operate independently due to magnetic coupling interference provided by, since coupling was eliminated between the flyback transformers.



(a) Conventional transformer in PLECS.



(b) Proposed transformer in PLECS.

Figure 5.14: Magnetic elements used in efficiency comparison.

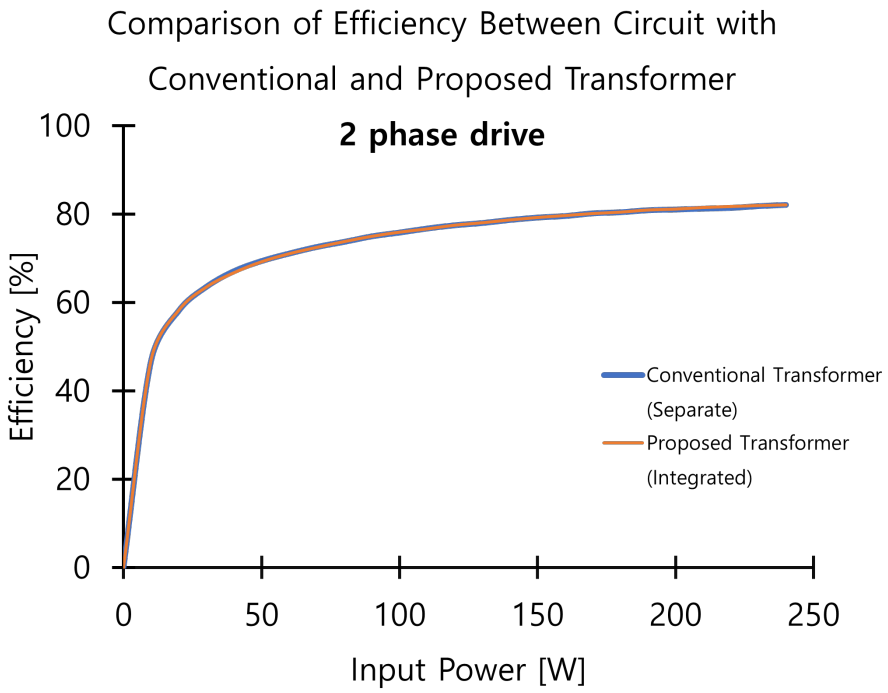
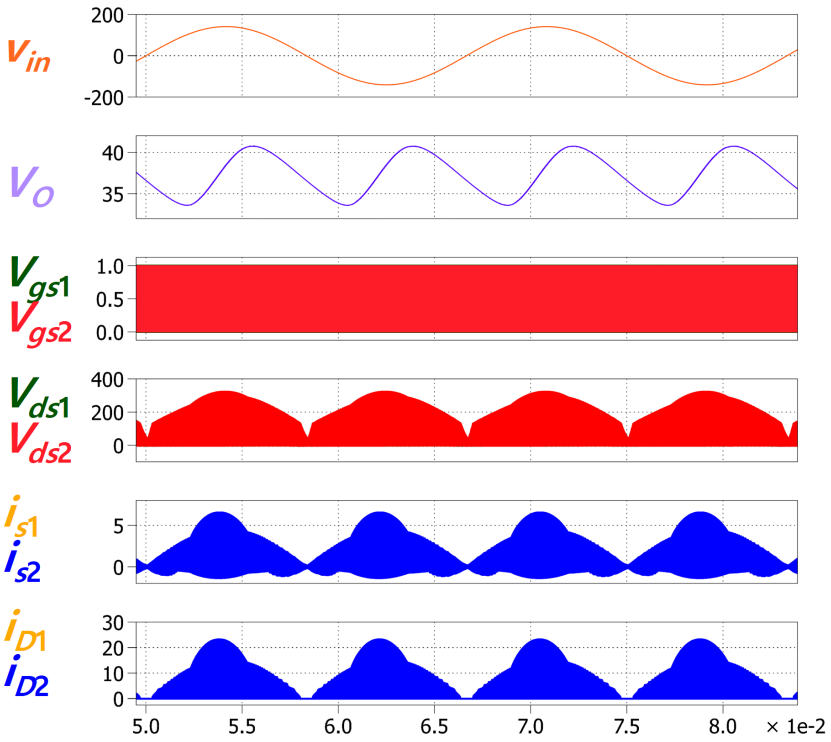
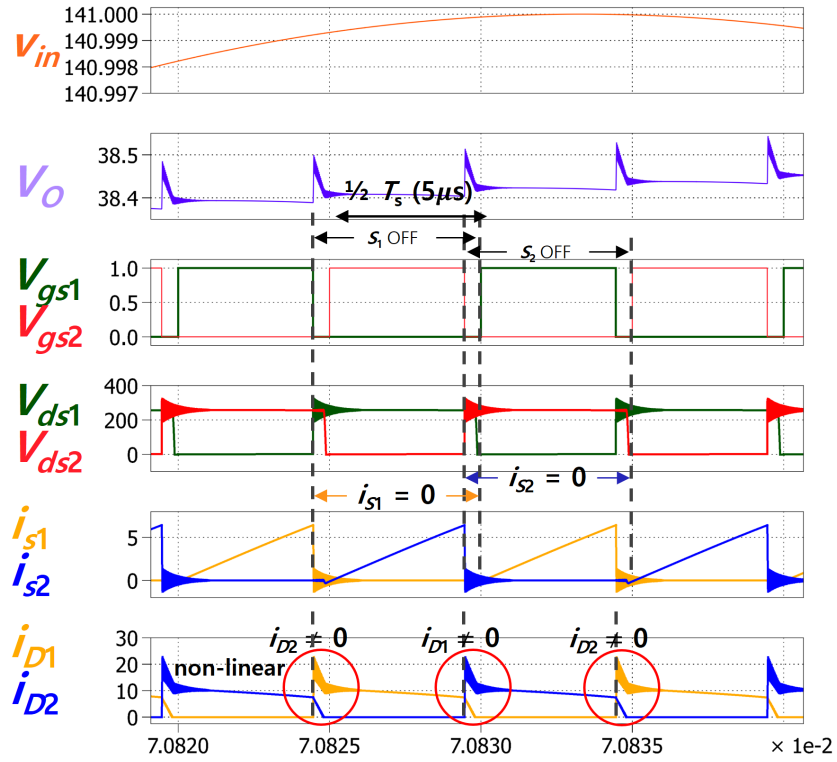


Figure 5.15: Comparison of efficiency between circuit with conventional and proposed transformer (2-phase drive).



(a) Zoomed-out simulated waveforms.



(b) Zoomed-in simulated waveforms.

Figure 5.16: Simulated waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at 120 W, one-phase AC drive, with $k_{N_{12}N_{34},N_{56}N_{78}} = 0.8$.

An analysis of the efficiency at 100 W with the presence of coupling was performed. The $k_{N_{12}N_{34},N_{56}N_{78}}$ was changed from $0.0 - 1.0$. From **??**, the presence of coupling has a negative effect on the efficiency of the transformer, therefore necessitating the need to have the transformers operate completely independent of each other to make power conversion as efficient as possible.

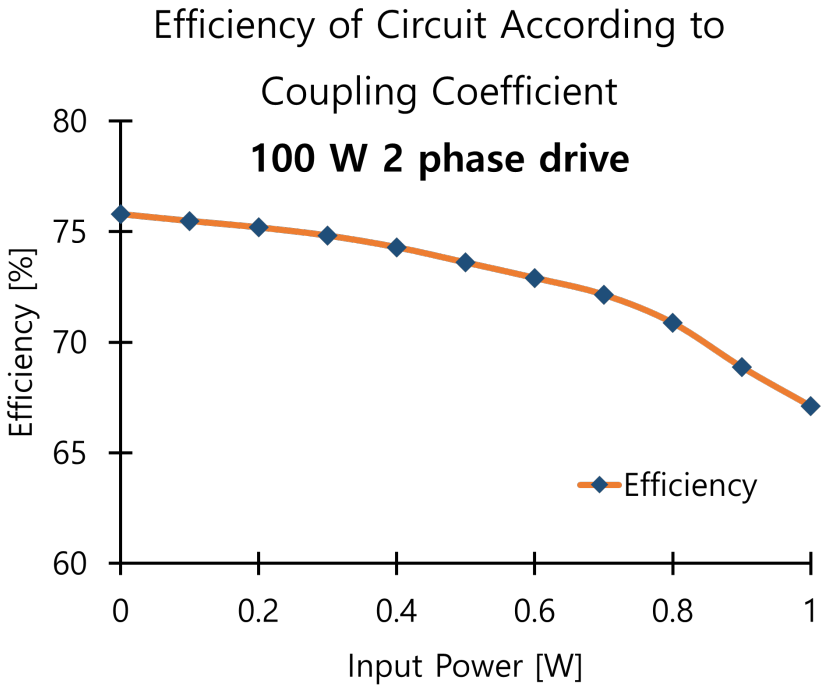


Figure 5.17: Circuit efficiency with respect to $k_{N_{12}N_{34},N_{56}N_{78}}$.

5.4 Experimental Verification

Since no control was present in this circuit, the duty had to be changed for each load level [put reference].

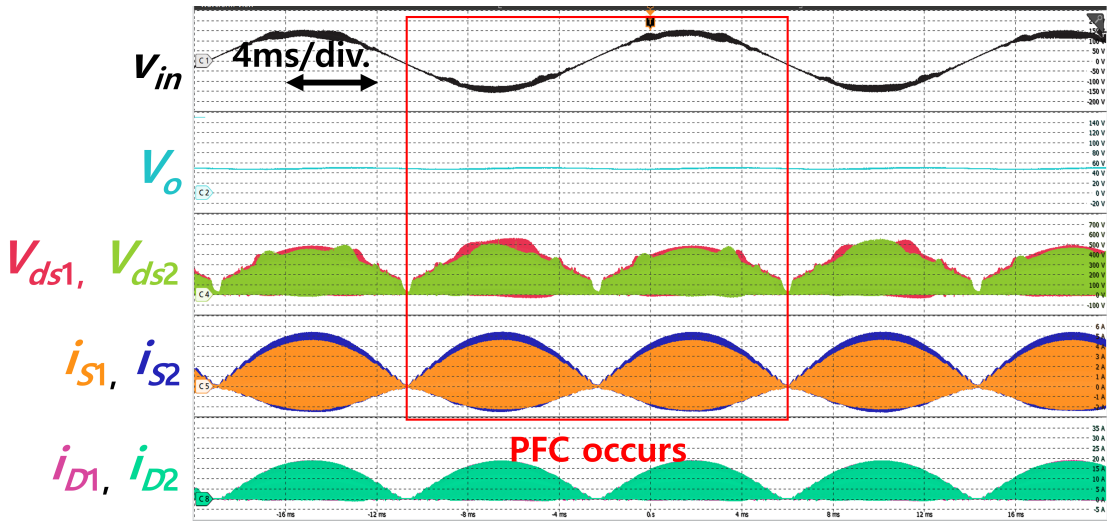
$$D = \frac{2}{V_{in}}\sqrt{\frac{0.5 L_m P_O}{T_s}} \tag{5.7}$$

This circuit was mainly created to verify the operation of the flux in the transformer.

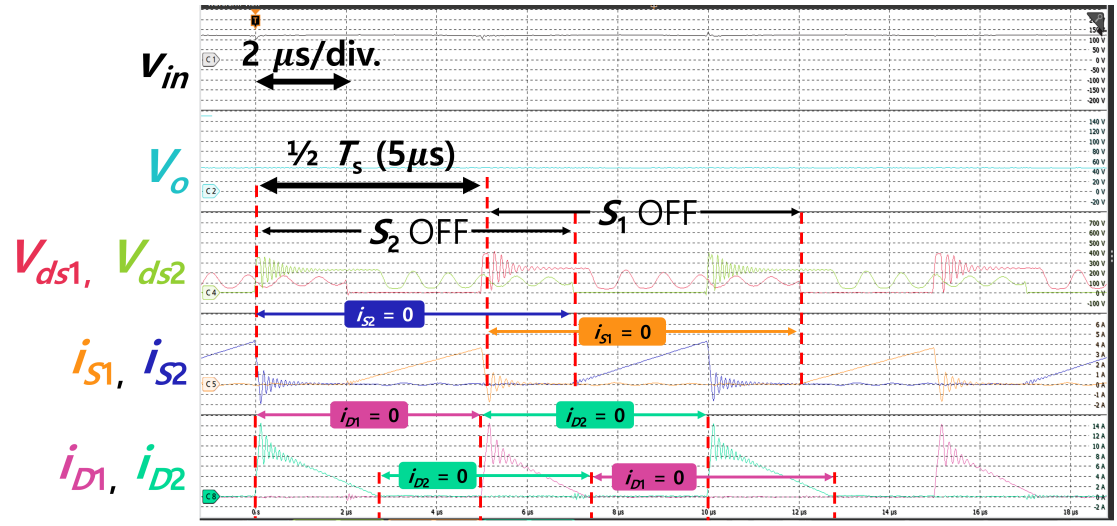
The board was tested in different operating conditions and different load conditions. For this study however, the circuit was tested at 100 V $V_{in,\text{rms}}$ and 48 V output voltage V_O, with 100 kHz switching frequency f_{sw}.

5.4.1 Two-phase AC Operation

Figure **??** shows the 100 W experimental waveforms at 4 ms/div. from the two-phase interleaved flyback rectifier with the proposed transformer, demonstrating power factor correction. Figure **??** presents 2 μs/div. waveforms, with the off-times of S_1 and S_2 indicated. The current i_{s_1} is zero during the off-time of S_1 and i_{s_2} is zero during the off-time of S_2. No current is induced in either switch except during the resonance between the leakage inductance and the parasitic capacitance of the switch. During the inactive period of the transformer, for example, between the start of the off-time of S_1 and the start of the off-time of S_2, the secondary diode side current is zero. Moreover, after i_{D1} and i_{D2} reached zero when S_1 and S_2 switched off, the currents remained at zero. Lastly, a linear increase and decrease of i_{S_1}, i_{S_2}, i_{D_1}, and i_{D2} was observed throughout the circuit operation. Therefore, from experiment, the two flyback converters N_{12}, N_{34}, S_1, D_1, C_O, V_O and N_{56}, N_{78}, S_2, D_2, C_O, V_O can operate independently without magnetic coupling interference, as coupling was eliminated between the flyback transformers.



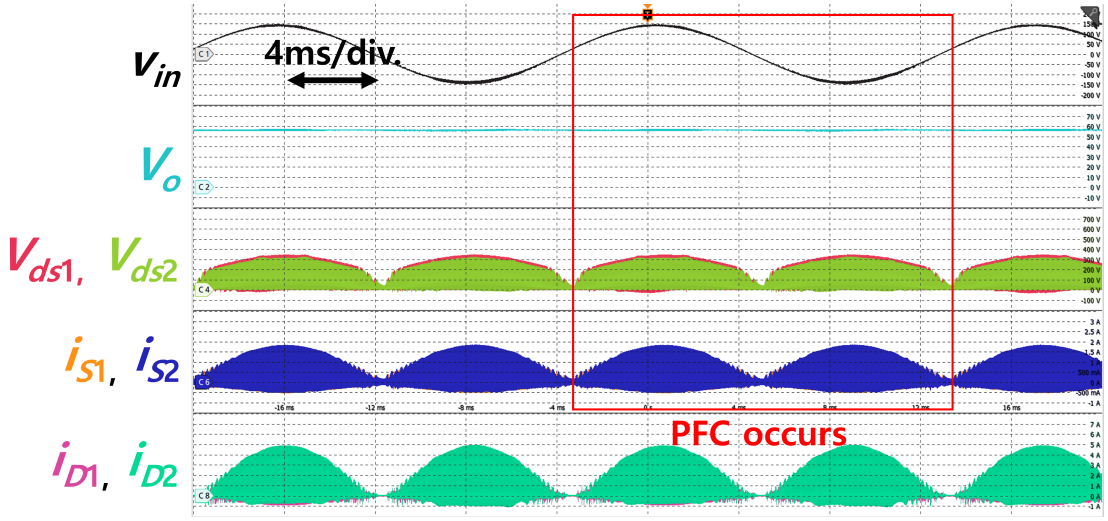
(a) Zoomed-out experimental waveforms



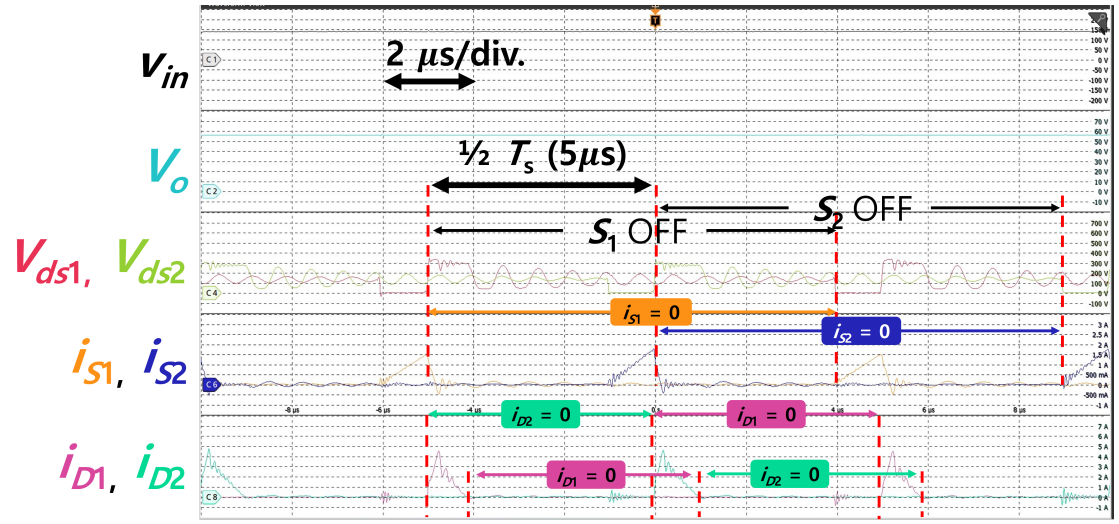
(b) Zoomed-in experimental waveforms.

Figure 5.18: Experimental waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at 100 W, two-phase AC drive. Here, v_in: 50 V/div.; V_o: 20 V/div., v_{ds_1}, v_{ds_2}: 100 V/div., i_{S_1}, i_{S_2}: 1 A/div., i_{D_1}, i_{D_2}: 5 A/div.; v_{ds} stands for drain-source voltage.

The same phenomenon can be seen at low (10 W) and high loads (240 W), which therefore implies that the transformer can operate separately in two phases at any load with out any magnetic interference. The results are shown in Figure **??** and **??**. However, as clearly seen from the result at 240 W (Figure **??**), there are are a few points of improvement for the current state of the converter and transformer. For instance, the peak values of the switch currents are not the same for each phase, which therefore means that the unequal value of magnetic inductances affect the operation of the circuit. Moreover, from examining the waveforms of i_D, the discharge time has also been affected. For a set peak current and constant output voltage, a change in L_m will affect the discharge time. A bigger magnetizing inductance means a longer discharge time. This can be solved by ensuring precise manufacturing and core selection for the transformer, such that the magnetizing inductances of the two transformers can be easily set to be equal to each other.

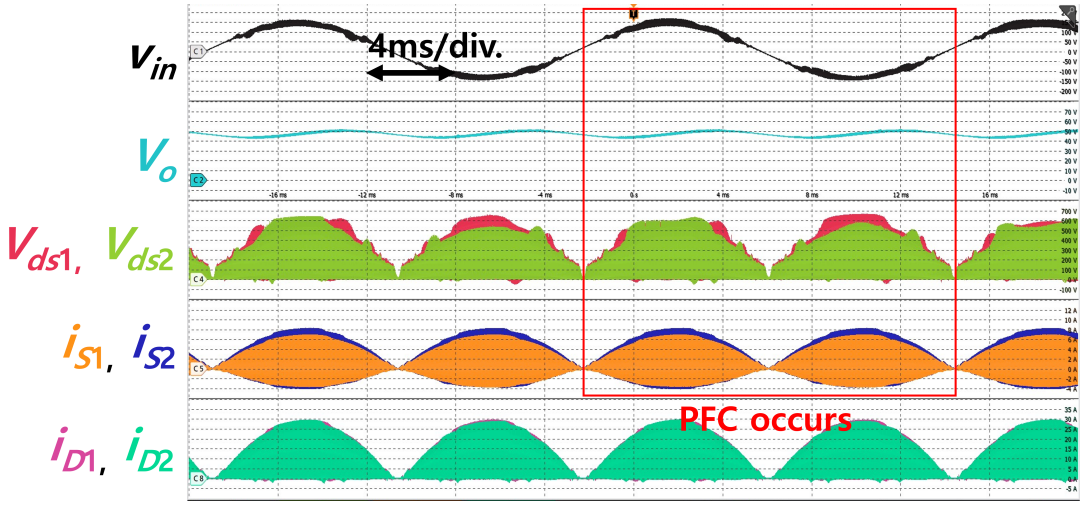


(a) Zoomed-out experimental waveforms.

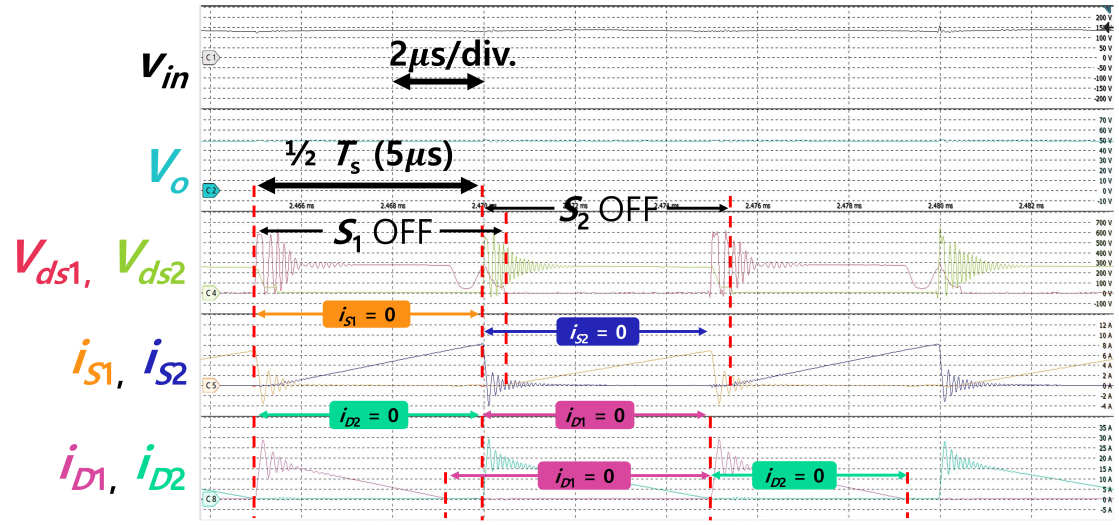


(b) Zoomed-in experimental waveforms.

Figure 5.19: Experimental waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at 10 W, two-phase AC drive. Here, v_in: 50 V/div.; V_o: 10 V/div., v_{ds_1}, v_{ds_2}: 100 V/div., i_{S_1}, i_{S_2}: 500 mA/div., i_{D_1}, i_{D_2}: 1 A/div.; v_{ds} stands for drain-source voltage.



(a) Zoomed-out experimental waveforms.

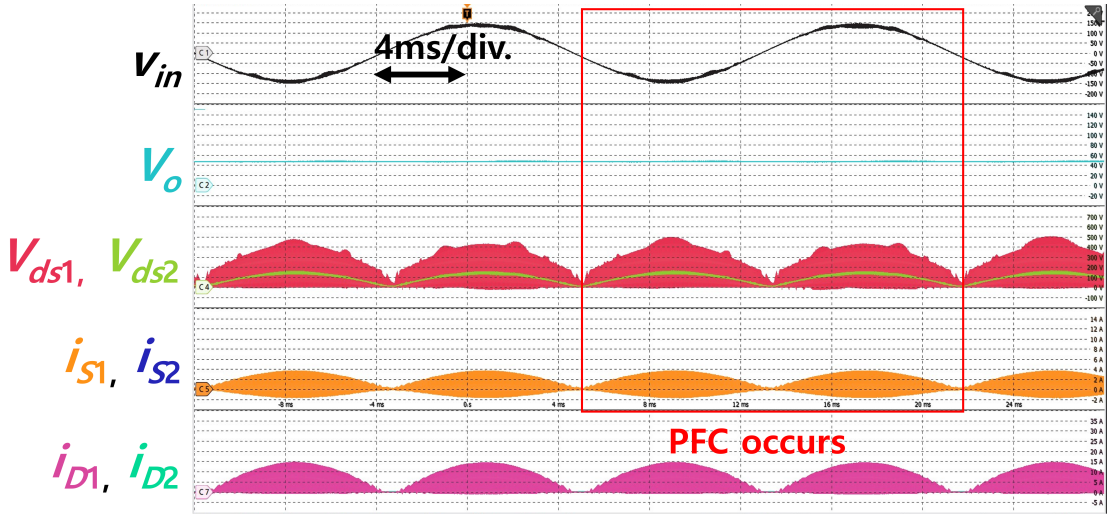


(b) Zoomed-in experimental waveforms.

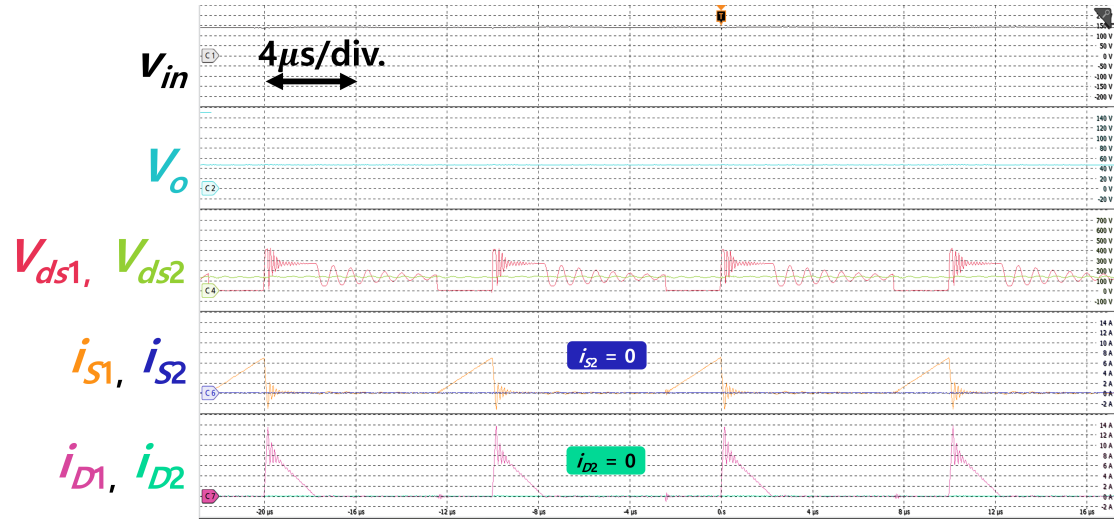
Figure 5.20: Experimental waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at 240 W, two-phase AC drive. Here, v_in: 50 V/div.; V_o: 10 V/div., v_{ds_1}, v_{ds_2}: 100 V/div., i_{S_1}, i_{S_2}: 2 A/div., i_{D_1}, i_{D_2}: 5 A/div.; v_{ds} stands for drain-source voltage.

5.4.2 Single-phase Operation

Single-phase drive at lower power is also feasible with this transformer. Figure **??** shows the 32.5 W, single-phase drive of S_1, 4 ms/div. waveforms two-phase interleaved flyback rectifier with the proposed transformer, demonstrating power factor correction. Figure **??** presents 2 μs/div. waveforms. The currents i_{S_2} and i_{D_2} are zero all throughout the circuit operation. This then confirms that only the flyback converter N_{12}, N_{34}, S_1, D_1, C_O, and V_O are operating. There is a voltage seen at V_{DS_2} since the two flyback converters are in parallel, although the switch is kept off. Therefore, from simulation, non-influence of the inactive transformer to the active one is confirmed.



(a) Zoomed-out experimental waveforms.



(b) Zoomed-in experimental waveforms.

Figure 5.21: Experimental waveforms of a two-phase interleaved flyback rectifier with the proposed integrated transformer at 32.5 W, one-phase AC drive of S_1. Here, v_in: 50 V/div.; V_o: 20 V/div., v_{ds_1}, v_{ds_2}: 100 V/div., i_{S_1}, i_{S_2}: 2 A/div., i_{D_1}, i_{D_2}: 2 A/div.; v_{ds} stands for drain-source voltage.

5.4.3 Converter Efficiency

The experimental efficiency has been obtained and the results are shown in Figure **??**. Converter efficiency continuously increases with load variation. Low efficiency is expected for light loads. At the highest load (240 W), the efficiency is around 85%. To further improve the efficiency, better components and circuit design can be used to lessen component losses. Moreover, control can also be added. Lastly, as had been explained, loss analysis is going to be performed to understand more the behavior of the transformer according to the circuit specifications and be able to optimize the design of the transformer more.

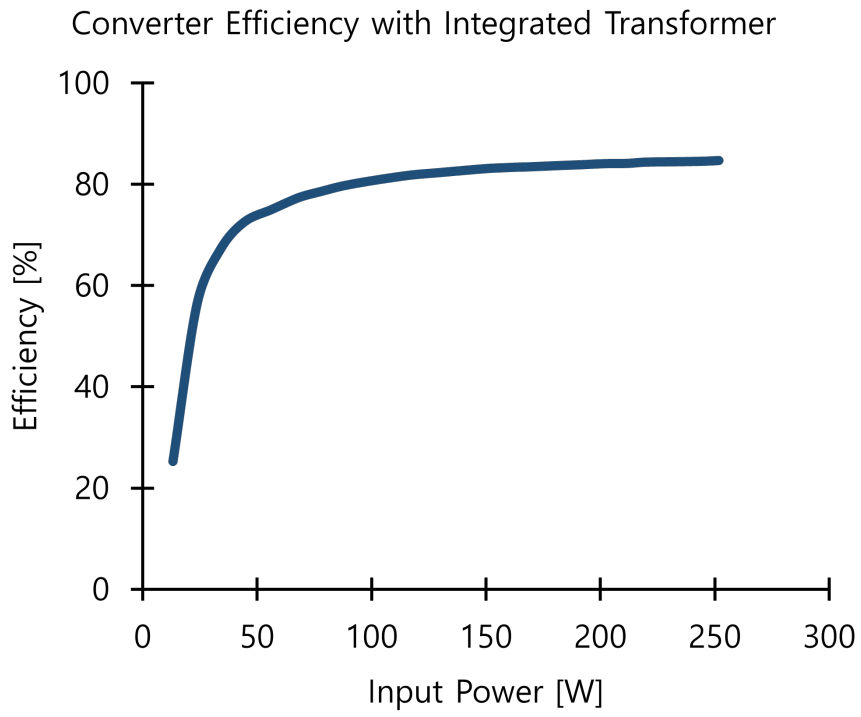


Figure 5.22: Experimental efficiency of circuit with proposed transformer (2-phase drive)

Chapter 6

Summary and Conclusion

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Appendix A

Appendix Title

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